Self-supervised generation of realistic training data enables nanoscale localization in challenging conditions

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Abstract

Localization microscopy has overcome the diffraction limit, i.e. the conventional resolution limit of a microscope, enabling nanoscale biological imaging by precisely determining the positions of individual emitters such as single fluorescent molecules. However, the performance of deep learning methods, commonly applied to these tasks, depends significantly on the quality of training data, typically generated through simulation. Creating simulations that perfectly replicate experimental conditions remains challenging, resulting in a persistent simulation-to-experiment gap. To bridge this gap, we propose a physics-informed generative model leveraging self-supervised learning directly on experimental data. Our model extends the Deep Latent Particles (DLP) framework by incorporating a physical model of the Point Spread Function (PSF; the image of a single point source in the microscope) into the decoder, enabling it to disentangle learned realistic environments from emitters. Trained directly on unlabeled experimental images, our model intrinsically captures realistic background, noise patterns, and emitter characteristics. The decoder thus acts as a high-fidelity generator, producing fully labeled, realistic training images with known emitter locations. Using these generated datasets significantly improves the performance of supervised localization networks, particularly in challenging scenarios such as complex backgrounds and low signal-to-noise ratios. We demonstrate our approach on a variety of experimentally measured microscopy data, including super-resolution imaging in 2D and 3D and particle tracking in live cells, showing substantial improvements in localization precision and emitter detection.

Keywords: Simulation-to-experiment gap · Single-molecule localization microscopy · Physics-informed generative model · Self-supervised learning

Introduction

Determining the precise positions of single molecules or particles in optical microscopy images is a fundamental capability enabling insights at the nanoscale, with applications ranging from biology to materials science. Single-molecule localization microscopy (SMLM) techniques, including PALM[1], STORM[2], and PAINT[3], use sequential localization of emitters to reconstruct images that overcome the diffraction limit, allowing detailed biological visualization[4]. Single-Particle Tracking (SPT) utilizes localization over time to determine the trajectories of moving molecules or nanoparticles, revealing information about their dynamics, interactions, and microenvironments[5]. Extending localization microscopy to three dimensions (3D) typically requires physical modification of the microscope; a common approach involves Point Spread Function (PSF) engineering, where optical elements modify the PSF to encode depth[6–9] or other information, such as spectral properties[10,11] or molecular orientation[12,13].

In localization microscopy, each emitter, e.g. a fluorescent molecule or a nanoscale particle, appears as a PSF, namely, the image of a point source formed by the microscope. In 2D SMLM the PSF is the standard, approximately Gaussian diffraction-limited spot, and localization amounts to estimating its lateral (x, y) center. Recovering the axial (z) position can be done based on PSF engineering , such as the optimally photon-efficient Tetrapod PSF[8,9] (Figure 1a). Figure 1b shows an example application of 3D localization: fluorescently labeled DNA loci in fixed yeast cells, where each cell contains a single locus whose Tetrapod-shaped image encodes its 3D position. The goal here would be to determine the 3D positions of all loci in the field of view from a single 2D image.

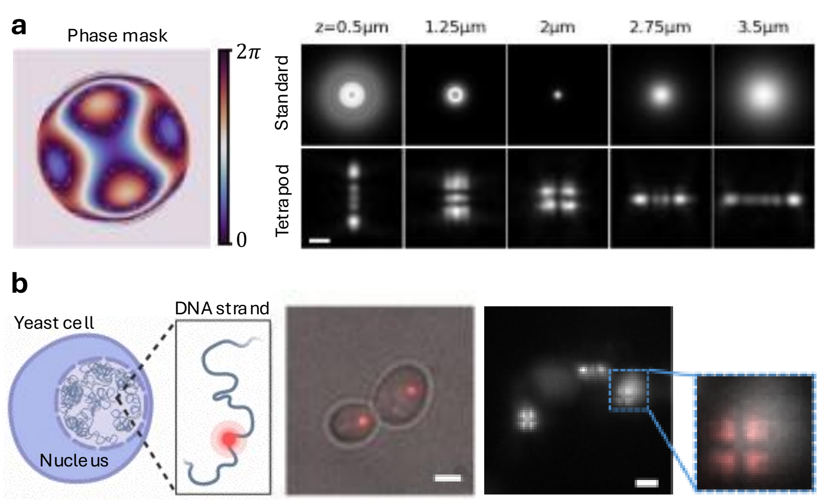

Figure 1: (a) Left: Model of a Tetrapod phase mask placed in the microscope's pupil plane encodes the emitter's axial position in the shape of the PSF. Right: measured Tetrapod PSF as a function of emitter axial position z, compared to the standard diffraction-limited PSF. Scale bar, 1 μm. (b) Biological sample: fluorescently labeled DNA loci in fixed yeast cells, where each cell contains a single labeled locus within the 3D volume of the nucleus (left: illustration; middle: brightfield overlay, Scale bar 1 μm). Adapted from ref [14]. The fluorescence image (right, Scale bar 2 μm) shows Tetrapod PSFs encoding the 3D positions of several emitters (inset: a single locus).

Deep learning methods have become essential tools in microscopy and SMLM, enabling powerful analysis and reconstruction[15,16]. They have been applied to various challenges, including the localization of dense emitters[17–19], the classification of emitter color[20], the determination of molecular orientation[21–23], and the facilitation of end-to-end optimization of imaging systems, e.g. designing the PSF together with the reconstruction algorithm[14,18,20,24]. Importantly, the performance of deep learning approaches, particularly supervised methods, depends on the quality and quantity of the training data. Such training data is often generated via simulation, which, in the case of localization microscopy, relies on modeling the PSF and other experimental factors, because acquiring large, accurately annotated experimental datasets for 3D localization microscopy is practically impossible.

However, building simulations that perfectly mimic experimental conditions is a major challenge[25]. Accurately capturing complex, spatially varying background, realistic noise statistics, and system-specific aberrations requires significant effort and in many cases is impractical. This leads to a persistent simulation-to-experiment (sim2exp) gap, where models trained on synthetic data underperform when applied to real measurements. Furthermore, even when an adequate simulation model exists, modifying it to reflect new experimental conditions can require time-consuming manual tuning or substantial redevelopment efforts.

Recent large vision foundation models, including adaptations for microscopy[26–28], show promise for tasks such as segmentation, cellular mapping or classification, but are ill-suited to sub-pixel localization: they are usually pre-trained on different domains and cannot be readily fine-tuned for SMLM given the scarcity of labeled data, especially across diverse SNRs and engineered PSFs. In the domain of data generation, diffusion models were adapted to synthesize realistic super-resolution images for augmentation[29], but the generated data is typically unlabeled, and thus cannot be used for training localization networks. Other generative approaches, such as Generative Adversarial Networks (GANs)[30–33], also attempt to bridge the sim2exp gap by learning mappings from simulated to real domains or generating images from labels. However, such post-hoc image transformations are fundamentally incompatible with the sub-pixel precision demands of SMLM. Any pixel-level modification of a simulated image can deform the PSF shape or shift its apparent center, breaking the correspondence between the image content and the original ground-truth emitter coordinates. At nanoscale resolution, where localization errors on the order of tens of nanometers are meaningful, there is no tolerance for label corruption introduced by an unconstrained generative process. Concurrently, VAE models demonstrated the ability to encode emitters' z-position in their latent space[34].

Here, we introduce a physics-informed VAE that embeds an image formation model directly in the generator to produce realistic, label-consistent localization-microscopy data. We propose *PILPEL – Physics-Informed Latent Particles for Emitter Localization*, an object-centric self-supervised generative model built on the Deep Latent Particles (DLP[35,36]) framework, to bridge the sim2exp gap. We extend DLP by incorporating an explicit physical PSF model into the decoder that takes encoded spatial coordinates as input. Trained directly on unlabeled experimental images, this architecture allows the model to intrinsically learn realistic background and noise patterns while disentangling the learned environment from the precise placement and properties of the emitters. The trained PILPEL model thus serves as a data generator, producing realistic, fully labeled training images with known ground truth emitter locations. This generated data is then used as highly realistic training material for supervised localization algorithms such as DeepSTORM[17] and DeepSTORM3D[18]. We demonstrate that using these generated datasets significantly improves the performance of supervised localization networks on real-world data, particularly in challenging scenarios such as complex backgrounds and low signal-to-noise ratios. No manual tuning of simulation parameters or explicit modeling of experimental attributes is required.

Results

Our goal is to design a controllable generative model that can generate realistic image data for the task of SMLM. In the following, we describe our proposed model that extends the DLP framework to localization microscopy.

Physics-Informed Deep Latent Particle Model for Localization Microscopy

DLP[35,36] is a self-supervised, object-centric image representation model, built on a variational autoencoder: an image is encoded into a set of latent "particles," each carrying features such as 2D position and size, together with a separate background representation. These are then decoded and combined to reconstruct the input (see Methods). Training on this image-reconstruction objective, with no labels, disentangles the particles from the background. In localization microscopy, the objects of interest are individual emitters, each represented by a PSF (Figure 1). Precise localization of these PSFs is the basis for the nanoscale resolution achieved in SMLM. DLP's explicit, position-based architecture allows fine-grained control over the objects in the generated output, making it a natural fit for our goal. We adapt DLP for localization microscopy by enabling it to recover 2D or 3D positions and intensities of point-emitters, observed as individual PSFs. We describe the model in its general 3D form; 2D localization follows as the special case in which the axial coordinate is fixed (z = 0) and the engineered phase mask is replaced by a flat-phase pupil, reducing the PSF to the standard diffraction-limited spot.

We extend the original DLP model to explicitly encode the key emitter properties, i.e., spatial coordinates ($\{x, y, z\}$) and intensity (I), such that each latent particle corresponds to a single emitter. Crucially, the model learns additional image components such as local and global background, SNR, and effective emitter density. We term this extended framework **PILPEL**: **P**hysics-**I**nformed **L**atent **P**articles for **E**mitter **L**ocalization.

This multi-component learning is achieved by incorporating a physical model of the PSF into the decoder, where the encoded $\{x, y, z\}$ coordinates are used as input to simulate PSFs as part of the image formation process. The PSF model serves as a strong inductive prior, embedding physical constraints into the generative process. The resulting physics-informed architecture allows the model to disentangle the PSFs from their surrounding scene, including noise and measurement artifacts, while explicitly capturing the key parameters of interest, namely, emitter location and intensity, enabling fine-grained control and accurate reconstruction. The framework is agnostic to the specific PSF: adapting it to a different engineered PSF (e.g. astigmatism[6], double-helix[7]) is straightforward and requires no architectural changes.

Our model architecture is illustrated in Figure 2: an object-centric encoder–decoder in which a differentiable physical PSF model, embedded in the decoder, drives image formation from the latent emitter coordinates. The reconstructed image is composed of separable components, namely, emitter PSFs, their local background, and a global background.

Once trained on real measurements, the decoder of our PILPEL model serves as a high-fidelity generator of extremely realistic images with precisely known emitter positions (Figure 2, right). Importantly, because PILPEL is trained for image reconstruction rather than localization, its encoded coordinates are not precise enough to serve for super-resolved localization directly. Instead, we sample new, exactly known coordinates into the decoder's physical layer, which are the ground truth for the generated image by construction (see Methods for the full details). These generated, labeled images inherit the realistic background and noise characteristics learned from the experimental data, as well as emitter intensities, densities, and their spatial distribution. This enables large and accurately labeled training datasets for supervised localization models, specifically DeepSTORM[17] and DeepSTORM3D[18] in this work. This data closely mimics real measurements, thereby narrowing the sim2exp gap while automatically capturing unmodeled experimental nuances.

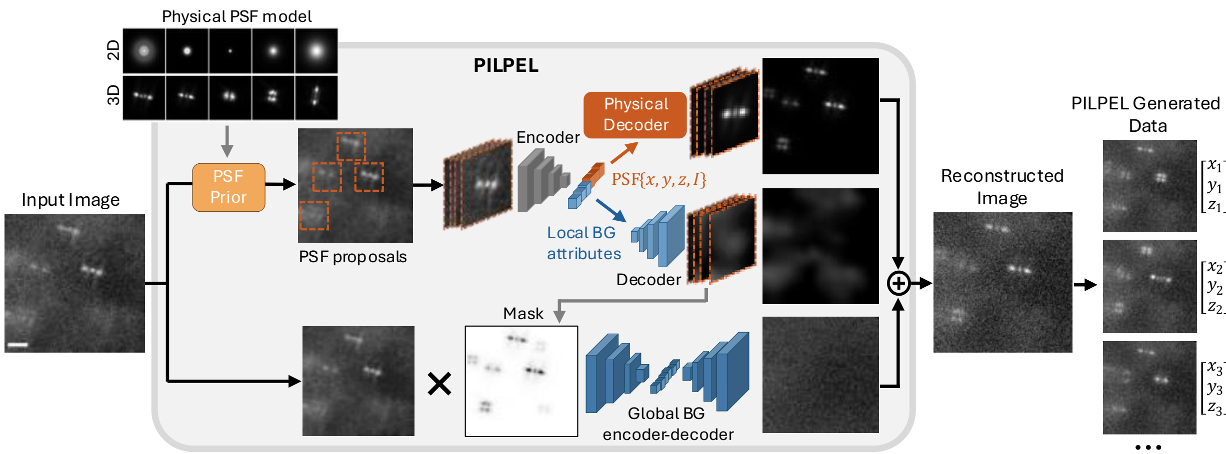

Figure 2: PILPEL – Physics-Informed Latent Particles for Emitter Localization. Encoding-decoding scheme with a latent representation of PSF-objects. The model divides the input image into overlapping patches and performs template matching using a precomputed PSF dictionary (standard 2D or engineered 3D PSFs, top left) to propose candidate emitter locations. Patches around these locations are then processed by an encoder that generates disentangled latent vectors encoding spatial coordinates, intensity, and appearance features of the local background. A decoder reconstructs the image using a physics-based PSF generator and a CNN-based local-background generator. These generated patches, along with a separately modeled global background, are combined to form the final reconstructed image. The entire system is trained end-to-end within a VAE framework, balancing reconstruction accuracy and latent regularization. Because the coordinates passed to the physical decoder are the ground truth of the generated image by construction, re-sampling them together with the latent attributes and passing them through the decoder provides new labeled images, with known {x, y, z} per emitter, that serve as training data for supervised localization networks. Scale bar, 2 µm.

In the following sections we show that the generation of realistic data improves the downstream performance of localization tasks. We first establish the method's accuracy on controlled simulations with known ground truth, then demonstrate its impact on real biological data, consisting of one dataset of 2D localization and three datasets of 3D localization. Full experimental details are provided in the Supplementary.

Bridging the sim2exp Gap - Controlled Simulations

To quantify performance and demonstrate the benefit of using training data generated using PILPEL, we evaluated DS3D for 3D localization across different training datasets. First, we designed a test set resembling yeast cell measurements with point sources shaped by a Tetrapod phase mask[37,38]. To simulate the typical autofluorescence observed in these cells, we added random asymmetrical 2D Gaussians around each PSF. Crucially, to reflect experimental conditions containing unknown noise profiles, we introduced complex patterns that are absent from the original DS3D simulator and are difficult to tune manually. Specifically, we simulated inhomogeneous backgrounds using Perlin noise, which is commonly employed in microscopy image analysis to model non-uniform illumination and other artifacts[19]. We term this test set PerlinData.

We then trained DS3D on three different datasets: (1) the original simulator with a wide range of SNR and background values; (2) the original simulator with an added local background (the same asymmetrical Gaussians used in the test set, practically introducing prior knowledge manually); and (3) PILPEL-generated dataset trained on the PerlinData images.

Figure 3a illustrates these datasets, showing that, as expected, the generated data most closely resembles the test set. PILPEL successfully learned Perlin noise with various levels and complex background patterns. These elements are not modeled in the original DS3D simulator and contribute to the sim2exp gap that can significantly degrade localization network performance. Consequently, DS3D trained on this generated data achieves superior performance, both in localization accuracy and in the number of detected emitters.

Figure 3b summarizes the performance gains of using the generated dataset under challenging conditions, split into PerlinData and low-SNR-PerlinData to show the benefit is even greater in low-SNR cases. Localization RMSE improves by 19 nm (a 28% improvement), and the Jaccard index (measuring the overlap between detected and ground truth localizations) doubles from 0.45 to 0.92, compared to the baseline. While introducing prior knowledge into synthetic training data improves performance (yeast background, red bar), training on generated data (PILPEL, green bar) is still significantly better, and also eliminates the need for incorporating known or assumed prior information in the DS3D data simulator.

To further evaluate the detection capability of the proposed model and its robustness against false positives, we performed a Precision-Recall (P-R) analysis on the PerlinData test set. Ground truth emitters were matched to localized coordinates using a strict lateral distance threshold of 100 nm, the detection confidence threshold varied from 0 to 800 (the range of output possibilities) to generate the curves. As shown in Figure 3c, there is a decisive performance gap between our method and the standard baseline. The baseline fails significantly on unmodeled noise, with precision collapsing after approximately 50% recall. In contrast, PILPEL maintains near-perfect precision (~1.0) up to ~95% recall. Notably, at a fixed high precision of 0.9, PILPEL yields approximately $2\times$ higher recall than the baseline. This advantage is even more pronounced in the challenging Low-SNR regime. Here, PILPEL maintains > 0.9 precision up to 75% recall, whereas the baseline fails (dropping below 0.9 precision) at only 15% recall. These results demonstrate the robustness of our approach to unmodeled noise, showing how PILPEL-generated data mitigates the sim2exp gap. See Supplementary Sec. B for extended details.

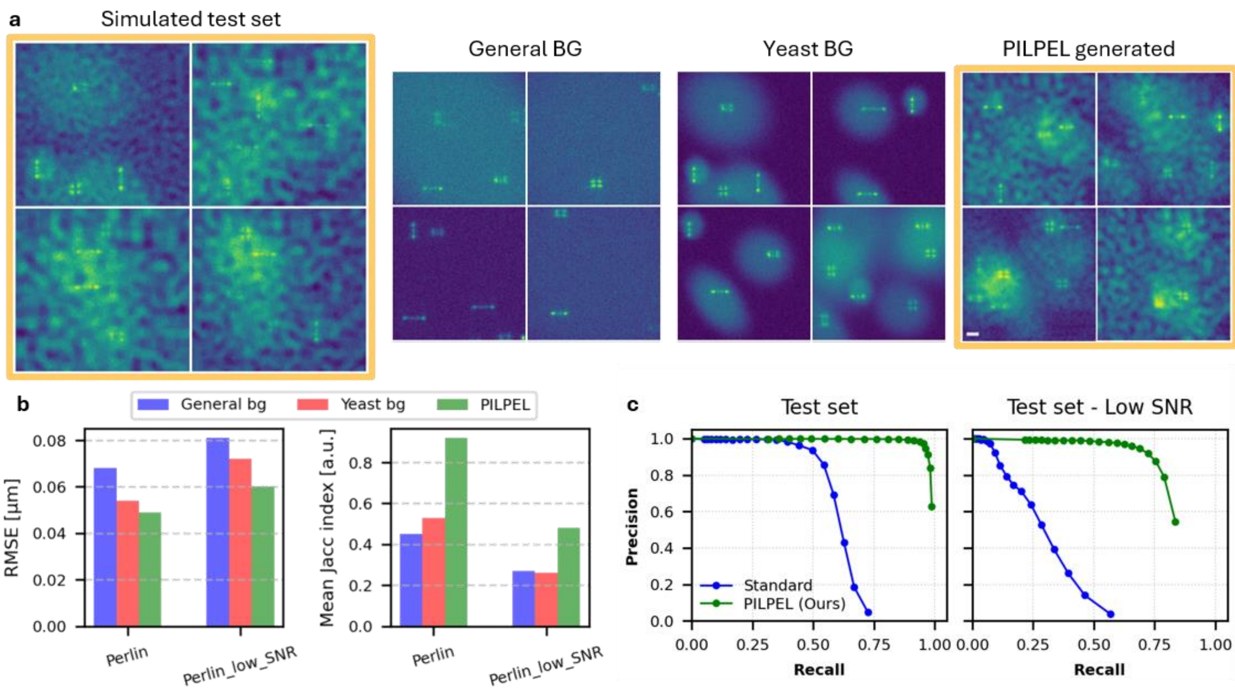

Figure 3: Robustness simulation – DS3D localization performance. (a) Comparison of synthetic training datasets used for DS3D localization of the Perlin test set (left). General BG – the standard background in DS3D; Yeast BG – includes local background around emitters to mimic yeast cells; and PILPEL – generated via our model. Scale bar, 2 μm. (b) Improvement of training on generated data. Localization RMSE (lower is better) and Jaccard index (higher is better) for DS3D trained on the generated dataset vs. training on yeast-BG and original-DS3D datasets. (c) Precision-Recall analysis on the PerlinData test set. PILPEL (green) demonstrates superior detection robustness compared to the standard DeepSTORM3D baseline (blue). While the baseline's precision degrades rapidly, particularly in the Low-SNR regime, PILPEL maintains high precision (> 0.9) across a significantly wider recall range.

2D Super-resolution Reconstruction of Microtubules

Having validated our approach under controlled conditions with known ground truth, we next evaluate its performance on real-world biological data, starting with 2D super-resolution imaging of microtubules, using DeepSTORM[17].

We used a dataset of microtubules in fixed cells, imaged via conventional 2D STORM[39]. We trained our PILPEL model directly on the experimental 2D STORM image frames. Since 2D localization requires only lateral coordinates, we fixed the axial coordinate in the PSF model to zero and used a flat pupil (zero phase mask), thus generating standard diffraction-limited 2D PSFs rather than the engineered 3D PSFs used in the other experiments. The model architecture remained unchanged, learning to represent each frame as a set of 2D point emitters with associated intensities, along with local and global background components. Figure 4a illustrates the model's learned decomposition of measured frames into their constituent components, demonstrating its ability to accurately represent the input data. The generated labeled dataset from PILPEL was then used to train DeepSTORM. We compared three reconstruction approaches on the same dataset: (1) ThunderSTORM[40], a classical localization algorithm based on sub-pixel fitting; (2) DeepSTORM trained using the original simulator provided with the published method; and (3) DeepSTORM trained on our PILPEL-generated dataset.

Figure 4b presents the super-resolution reconstructions of a microtubule structure. The original DeepSTORM (trained on simulated data) achieves sharper reconstructions compared to ThunderSTORM, especially in dense regions with overlapping PSFs. However, it still underperforms in certain challenging areas (e.g., lower left region), typically corresponding to low SNR regions or areas with characteristics that differ substantially from the training data. DeepSTORM trained on PILPEL-generated data demonstrates significantly improved performance in these problematic areas while further enhancing the overall reconstruction, maintaining sharp, resolved details (see Figure 4b insets) and improving visibility, thus outperforming both ThunderSTORM and the original DeepSTORM trained on simulated data.

The total number of localizations increased from $\approx$168K for the original DeepSTORM to $\approx$329K for the PILPEL-trained version across 600 frames (530$\times$358 pixels), representing a 1.96-fold increase. This improvement translates to both enhanced spatial resolution and more complete structural representation of the microtubule network. To empirically validate these results, we used Fourier Ring Correlation (FRC), filament structural analysis, and spatial distribution analysis, as detailed in Supplementary Sec. C.

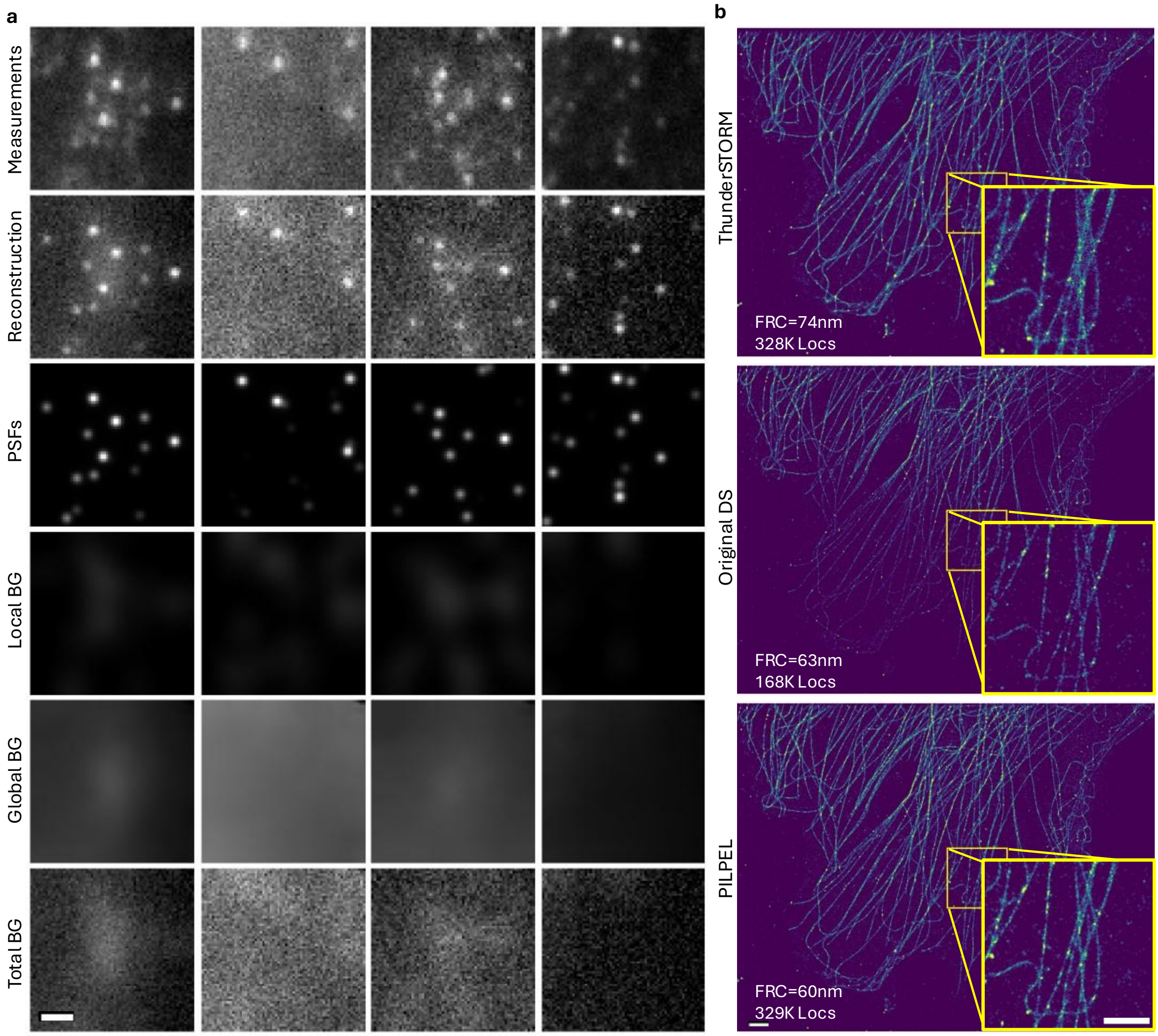

Figure 4: (a) *2D STORM experimental frames and PILPEL output components. The first row shows five measured images*, the second row shows PILPEL's final reconstructions. T*he remaining rows show PILPEL's reconstruction components:* the PSF component (third row) representing the isolated emitter signals; the local background component (fourth row) around each emitter; the global background component (fifth row), representing the general background across the entire field of view; and the total background which is the sum of all non-PSF components, including the added noise. *Scale bar, 2 µm.* (b) 2D super-resolution reconstruction comparison of ThunderSTORM, original DS, and DS trained on PILPEL-generated data. Compared to ThunderSTORM, original DS better recovers dense structures, and PILPEL-trained DS further enhances the reconstruction. Scale bars, 3 µm (main), 2 µm (inset).

3D Super-resolution Imaging of Mitochondria

Having demonstrated PILPEL in the 2D setting, we now turn to the more challenging 3D case, where depth is encoded through an engineered PSF. We apply the method to a 3D super-resolution dataset of mitochondria[41] imaged with the Tetrapod PSF. As shown in Figure 5a, training on our generated data dramatically improves the final reconstruction. The number of detected localizations increased from 119K to 581K ($\sim 4.9\times$ increase) across 20K frames. This translates to improved image quality, i.e. fuller features (as seen in Figure 5b), and potentially faster acquisition. Figure 5c shows measured frames and corresponding PILPEL reconstructions.

A possible concern would be that an increase in detections may be associated with a higher rate of false positives. We evaluate this with a volumetric density analysis on 20 background ROIs compared to 20 volume-matched on-structure ROIs (Supplementary Sec. C). While PILPEL's off-structure density is indeed higher than DS3D (4.56 vs. 0.45 loc/μm^3) this absolute increase (4.11 loc/μm^3) is negligible compared to the signal gain on the biological structure (635.5 loc/μm^3). Furthermore, these residual localizations can be suppressed by applying a stricter confidence threshold, made possible by the higher total number of detections, or by using standard density filters commonly applied in SMLM reconstruction[40]. For the visualization in Figure 5, the reconstruction from our method was post-processed with such a filter to match the original's noise level, while the original reconstruction was left as-is, as filtering would degrade the sparse biological structure. Crucially, the high detection rate achieved by our method makes the application of such filters feasible. While our dense reconstructions survive filtering and preserve the underlying structure, the baseline's sparse detections vanish under the same processing. The unfiltered version is available in Supplementary Figure 8.

Importantly, our method learns directly from the measurements without the need for manual tuning of the DS3D simulation parameters, which requires hours if not days, depending on user expertise. Human capability to verify and check vast amounts of images is limited, whereas PILPEL learns to generate based on the entire dataset, providing more comprehensive optimization. Furthermore, manual tuning relies on subjective visual assessment ("does it look similar enough?"), while PILPEL employs computationally optimized image similarity loss between measurements and generated data, which is more efficient and robust. While recent work aims to automate this process[42], such approaches are still limited by the simulator's explicit model and pre-defined parameters, which do not capture unmodeled traits that our generative approach learns directly.

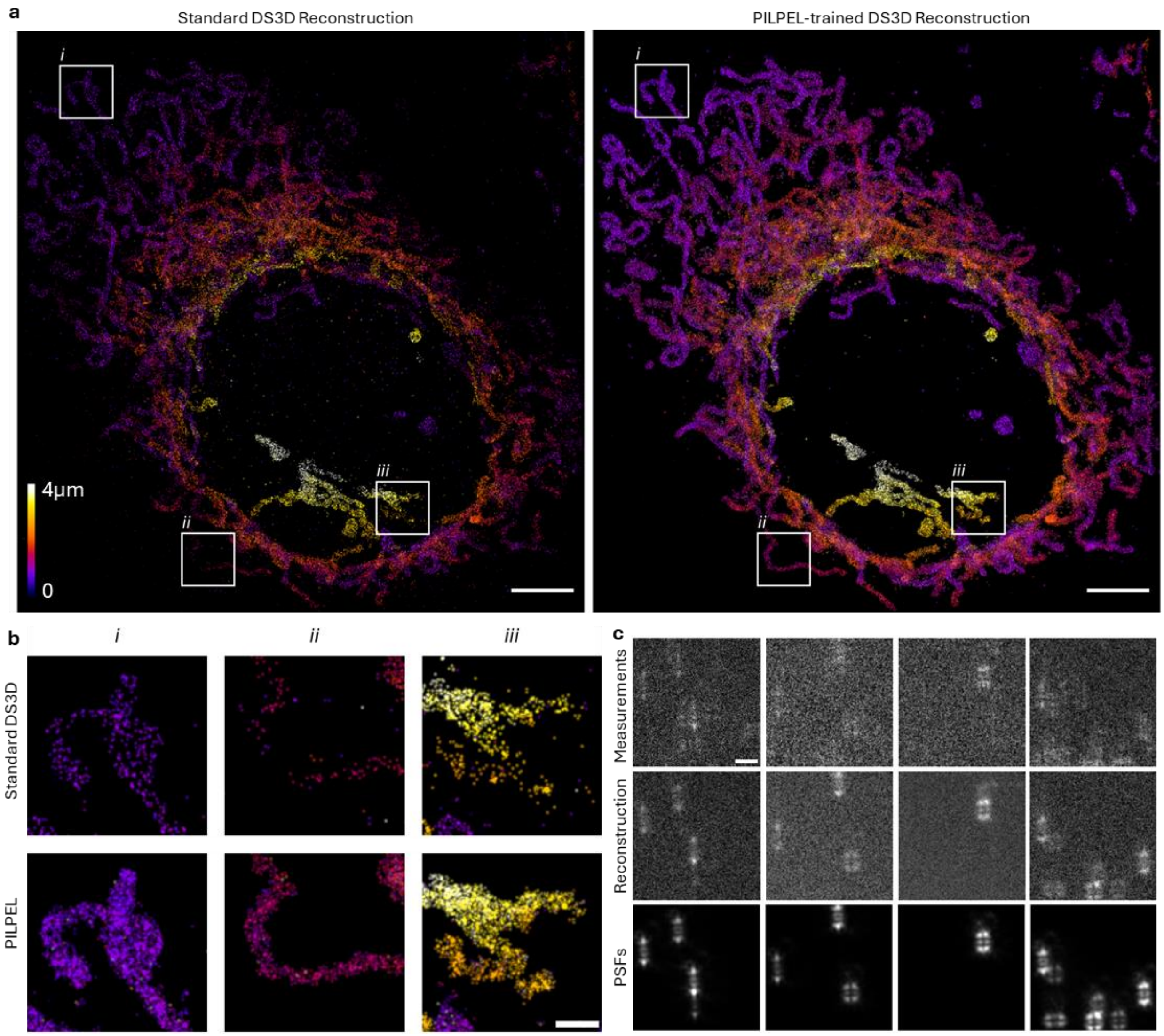

Figure 5: 3D super-resolution reconstruction of experimental data. (a) Super-resolved reconstructions of mitochondria (thickness ≈ 4 µm), rendered as 2D histograms with color-coded z-positions, using DS3D trained on the original simulated data (left) versus our generated data (right). Scale bar, 5 µm. (b) Magnified views of regions i, ii, and iii as indicated in (a). Scale bar, 1 µm. (c) Examples of randomly sampled measured images, accompanied with their matching PILPEL reconstruction generated by our model, used to train DS3D. Bottom row shows the PSFs component of the reconstructions, defined by $\{x, y, z\}$ coordinates, that serves as the GT training labels. Scale bar, 3 µm.

3D Localization of DNA Loci in Yeast Cells

Beyond super-resolution microscopy, we further assess the generalizability of our approach in a distinct biological system characterized by local background features not modeled in the original DS3D simulator. We apply the same pipeline to experimental images of fluorescently labeled

DNA loci in fixed yeast cells[38], where each cell exhibits a single fluorescent spot shaped by the Tetrapod phase mask. In such a biological system, an example application would be tracking loci over time in 3D under different conditions[5,38,43]; here, we use it to demonstrate our method under a noise regime distinct from STORM.

Dense low-SNR measurements were used as a challenging dataset for DS3D localization, comparing a model trained with the original simulator versus a model trained on generated data derived via our data generation pipeline. Figure 6a shows the measured dataset and the corresponding PILPEL reconstructions. Figure 6b displays the measured images alongside localization results from both localization networks. The general detection rate increased by $\sim 1.47\times$, from 12,561 detections with the original net to 18,461 detections using the PILPEL-generated dataset, across three time-lapse measurements (100 time points each) in a 1200×1200 FOV. In addition to that, mean detection confidence values increased by 40% using PILPEL.

Ground-truth positions are unavailable in such experiments, but standard GT estimation in single cells is MLE localization - fitting xyz coordinates of the PSF model together with asymmetric Gaussian background. This is relevant when the model matches the measurement, and MLE is not always correct in such difficult cases. Hence, we conducted a small experimental test set of 1,500 single-cell measurements with relatively high SNR where we assume the MLE fitting will work, and compared the results to the two networks' localizations. Localization RMSE improves from 107 nm to 96 nm, a 10% improvement, using PILPEL. Importantly - this test set comprised high SNR data, whereas a more significant improvement will be on low SNR cases, where the model deviates substantially from the measurement. The advantage of our method is that it generalizes the model and can identify more emitters, in a large FOV with multiple emitters. This is reflected in greater detection rate, seen also on the provided images, including low-SNR emitters missed by the original DS3D.

Compared to the synthetic yeast-BG dataset from Figure 3, our model's ability to learn unmodeled features is clearly demonstrated. While one can incorporate local background to the DS3D simulator, it still does not adequately resemble the actual measurements seen in Figure 6. PILPEL's generated data improves this correspondence and removes the need for manual implementation, allowing its use for future datasets with different noise patterns and shapes.

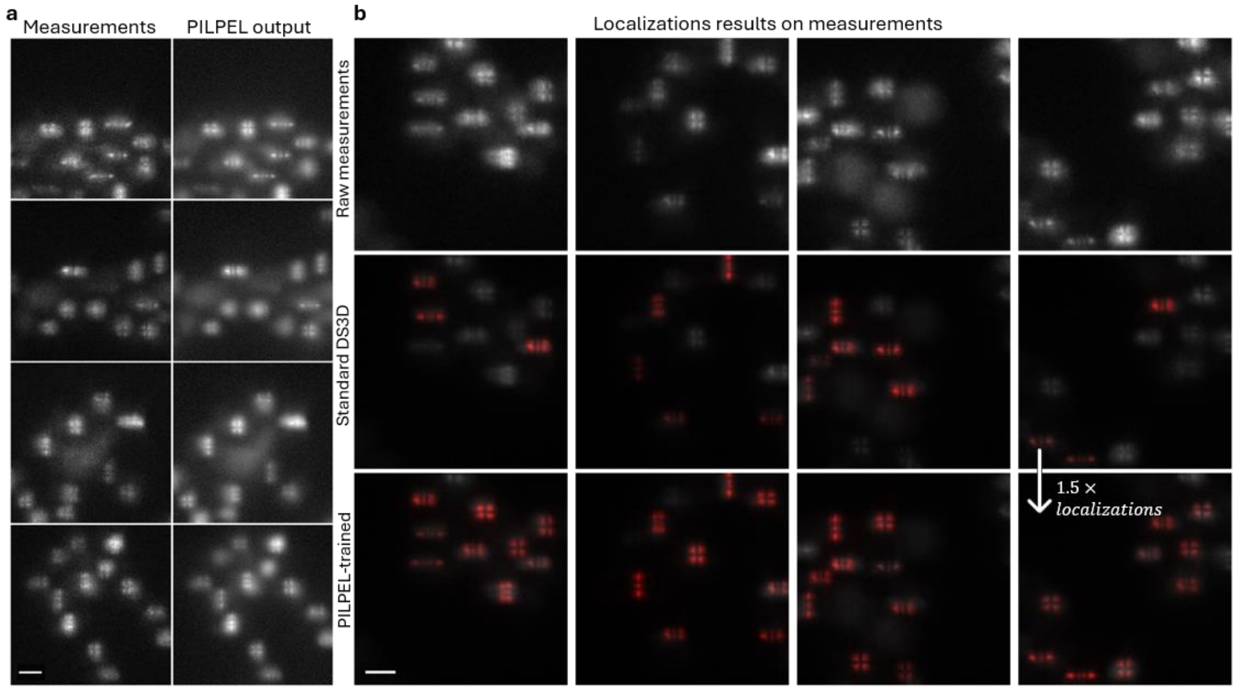

Figure 6: Fluorescently labeled DNA loci in yeast cells. (a) PILPEL trained on experimental data, showing dense yeast cells with pronounced, spatially varying local background. Left column is the input - raw measured images; on the right is the output – the reconstructions generated by the model. Reconstructions capture cell-specific background and noise patterns, illustrating that the model learns biological structures directly from the data. (b) Comparison of localization results. Localization overlays on the same measurements (white) with 3D detections rendered (red). Middle row: DS3D trained on standard simulated data; bottom row: DS3D trained on PILPEL-generated data. Evidently, the PILPEL-trained network yields visibly higher red-on-white overlap and recovers more emitters missed by the standard baseline. Scale bars, 3 µm.

Discussion

High fidelity data simulation is crucial for all simulation-based learning. In this work we pushed data simulation in localization microscopy to the extreme realistic limit. This is achieved by integrating an explicit PSF model within a self-supervised, object-centric framework that disentangles emitters from complex background and noise. As our benchmarks demonstrate, training supervised models on this generated data significantly enhances localization accuracy and detection rates, especially for challenging or novel experimental conditions. Furthermore, this approach eliminates the need for manual simulation tuning and the explicit modeling of complex image elements, offering a robust, adaptable solution for improving deep learning performance in microscopy.

To further support the generalizability of our method, we applied PILPEL to an additional 3D dataset of bacteria using a different depth-encoding PSF (astigmatism). The results (Supplementary Sec. A) demonstrate the applicability of PILPEL to another distinct biological sample, background profile, and SMLM modality.

Despite the advances demonstrated, our method exhibits several limitations. Training the generative model requires additional computational time compared to training on synthetic datasets, although it significantly reduces manual human intervention (see Supplementary Sec. F for computational details). Additionally, accurate reconstruction still necessitates a PSF model. This is obtainable either by a numerical model or, better, from experimental calibration (see Supplementary Sec. D for robustness analysis).

Future work includes leveraging this generative capability for a variety of microscopy scenarios that are hard to simulate, e.g., particle flows, or living organisms, extending to dynamic tracking via DDLP for video[36], and exploring latent space manipulation for tailored dataset generation for downstream tasks (e.g., denoising). Another promising direction is to incorporate learned aberrations[44], enabling also the optimization of the PSF model itself.

Overall, PILPEL replaces hand-crafted simulation with a generative model that learns the experiment itself, producing realistic, labeled training data automatically, extending the applicability of localization microscopy to increasingly challenging experimental datasets.

Methods

Downstream localization networks

In this work we demonstrate the use of PILPEL-generated data to train supervised localization networks, DeepSTORM for 2D and DeepSTORM3D for 3D localization, both of which conventionally rely on simulated training data. DeepSTORM (DS[17]) is a deep convolutional neural network (CNN) method designed for high-density 2D SMLM that reconstructs super-resolved images from diffraction-limited frames of blinking emitters. DeepSTORM3D (DS3D[18]) is a CNN-based method for dense 3D SMLM over large axial ranges. It analyzes 2D images containing overlapping engineered PSFs (such as the depth-encoding Tetrapod PSF, Figure 1a) and outputs a 3D localization map. These supervised networks are trained entirely on simulated data, where large datasets are generated using a physics-based model of the microscope and PSF. These simulations incorporate realistic experimental variability, including different emitter densities, signal-to-noise ratios, background levels, and noise models. A significant step in the analysis pipeline is the generation of training data that resembles the experiment. This involves significant manual tuning and can be highly time-consuming even for an experienced user. Our model, among other contributions, alleviates this manual tuning completely.

Deep Latent Particles (DLP)

Deep Latent Particles (DLP[35,36])[*] is a self-supervised, object-centric image representation model that decomposes an input image $x \in \mathbb{R}^{H\times W\times C}$ into a set of K structured foreground latent particles $L = \{l_p, l_s, l_d, l_t, l_f\}_{k=0}^{K-1}$. Each particle encodes the following attributes: $l_p \in \mathbb{R}^2$ denotes the 2D position, $l_s \in \mathbb{R}^2$ the spatial scale (bounding box size), $l_d \in \mathbb{R}$ the appearance order, $l_t \in \mathbb{R}$ the

[*] In our work, we use the improved DLPv2[36] as our base model and refer to it simply as DLP.

transparency, and $l_f \in \mathbb{R}^m$ the latent visual features. A key component of DLP is the spatial prior, where particle positions are determined using a spatial softmax (SSM) operator applied to activation maps from a patch-based convolutional network. In addition to the K foreground particles, the model includes a single background particle $l_{\mathrm{bg}} \in \mathbb{R}^{m_{\mathrm{bg}}}$ to represent the background visual features. The values of K, m, and m_{bg} are model hyperparameters. To learn these attributes, the model decodes the latent particles into glimpses and composes them together with the background to reconstruct the input image. Training is performed end-to-end using a variational autoencoder (VAE[45]) framework, with the objective of maximizing the evidence lower bound (ELBO), combining image reconstruction loss with a KL-divergence regularization term. The disentangled latent representations learned by DLP have been successfully applied to a variety of downstream tasks, including video prediction[36], reinforcement learning[46], and imitation learning for robotics[47]. For a detailed overview of the model, we refer the reader to the original work[36].

PILPEL architecture and training

In our modified model, the physics-informed prior replaces the spatial softmax (SSM)-based prior used in the original DLP to propose keypoint locations. Specifically, we employ a physical model of the 3D PSF that accounts for the optical parameters used in the experiment. This PSF model can be theoretically derived or experimentally calibrated, for example, by using phase retrieval methods such as VIPR[48]. Similar to DS3D[18], our model includes a differentiable optical layer that simulates image formation from emitter coordinates to the final sensor image. The same PSF model is used both in the prior module to propose emitter locations and in the decoder to generate the PSFs of each reconstructed patch.

As illustrated in Figure 2, the input image is initially divided into overlapping patches. Each patch undergoes a 3D template matching process against a dictionary of pre-calculated PSFs spanning the expected axial z range (for 2D localization, the dictionary reduces to the single in-focus PSF and matching is purely lateral). This yields an initial estimation of potential PSF positions (l_x, l_y, l_z coordinates) within each patch. The maximum cross-correlation score is used as a confidence metric to filter out patches unlikely to contain any PSFs, resulting in a proposed set of latent particles.

Following this coarse-grained proposal stage, we extract image patches centered around each proposed PSF location. These patches, intended to capture both the PSF and its local surroundings, serve as input to a particle encoder module. The encoder outputs the latent attributes for each proposed object. These are disentangled by construction, as detailed below, with specific latent attributes representing subpixel offsets (l_{dx}, l_{dy}, l_{dz}) to refine the initial position estimate, the PSF's intensity l_I, and additional latent attributes encoding the appearance features of the patch. The intensity factor l_I allows the model to capture experiment-specific SNR and brightness distributions and enables an automatic pruning mechanism (by driving the factor to zero during optimization), which adapts to varying emitter densities by removing unnecessary proposals.

The decoder reconstructs the image from these latent representations and consists of two distinct components. The first is the differentiable physical layer that takes the refined spatial coordinates (initial estimate and the learned offset, $l_x + l_{dx}$, etc.) and the intensity factor from the latent vector

l_I to generate a PSF image. The second component is a CNN-based decoder which takes the remaining latent appearance attributes and generates a corresponding image patch representing the local background surrounding the PSF. These are combined and placed into the final reconstructed image at their corresponding global coordinates.

To learn global background characteristics, the generated object patches are used to mask the input image. This masked input is processed by a separate encoder-decoder network dedicated to modeling and reconstructing the general background component of the scene. An additional encoder learns a latent attribute l_n, used as the standard deviation of random noise added to the general background.

Finally, the generated components (PSFs, local background patches and general background) are added together to produce the final reconstructed image. The entire model is optimized end-to-end within the VAE framework by minimizing the ELBO - the reconstruction loss (typically MSE or a perceptual loss) between the output image and the original input image, and KL-divergence regularization term for the stochastic latent variables. An ablation study evaluating the contribution of each module is provided in Supplementary Sec. E.

Generating labeled training data

To produce labeled training data, we encode an input image, sample new PSF coordinates (drawn randomly around the original encoded coordinates), and pass them through the decoder to generate realistic perturbations of the input image (Supplementary Figure 9). The model reproduces the global and local background characteristics learned from the real image, while the newly assigned PSF coordinates serve as the training labels. These synthetic yet realistic images, paired with their exact emitter positions, are then used to train a supervised localization network.

Notably, since PILPEL's objective is image similarity rather than localization accuracy, the encoded coordinates are not guaranteed to reach the sub-pixel precision required for single-molecule localization: the reconstruction is visually faithful to the input, but the encoded PSF positions may be slightly shifted from the true ones, with a discrepancy of approximately 90 nm 3D-distance RMSE in simulation - too large to use directly as ground-truth labels. However, this potential inaccuracy is irrelevant for our purpose, since the decoder is driven by new, precisely known coordinates fed into its physical layer, which by definition constitute the ground truth for the newly generated image.

Experimental datasets, sample preparation, and imaging

2D STORM (microtubules). This dataset, originally published in Saguy et al.[39] consist of fixed COS7 cells immunolabeled for α- and β-tubulin (Alexa Fluor 647), imaged by STORM with an oil-immersion objective (NA = 1.49, 100× magnification), as described in details in ref. [39].

3D STORM (mitochondria). This dataset, originally presented in Orange-Kedem et al.[41], consists of 3D STORM imaging of mitochondria in fixed COS7 cells labeled with Alexa Fluor 647. Imaging used a 4f Fourier-plane extension equipped with a Tetrapod phase mask encoding an axial range of 4 µm, and an oil-immersion objective (NA = 1.35, 100× magnification). The experimental PSF was retrieved with VIPR[48] from a calibration-bead sample prior to imaging. See ref. [41] for further details.

3D localization (yeast DNA loci). We imaged fluorescently labeled DNA loci in fixed *Saccharomyces cerevisiae* yeast cells. The cells were tagged with mCherry, generating red fluorescent spots at the *POA1* DNA loci (chromosome II); each cell exhibits a single fluorescent spot[38]. Imaging used a 4f extension to the optical system (equipped with an oil-immersion objective, NA = 1.49, 100× magnification), incorporating a Tetrapod phase mask designed for an axial range of 3 µm. The PSF model was retrieved using VIPR[48] from a calibration bead. Two datasets were acquired from different fields of view and experimental stages: one with high SNR and relatively sparse cells, and one with low SNR and higher density (including naturally denser FOVs resulting from cell aggregation, acquired later after some photobleaching). The dense, low-SNR dataset served as a challenging dataset for DS3D localization; the high-SNR dataset served as the test set for the localization-precision estimate reported in the main text.

3D localization (bacteria). This dataset, from Karempudi et al.[49], consists of 3D imaging of chromosomal loci labeled with YFP in live *E. coli*, using an oil-immersion objective (NA = 1.45, 100× magnification) with a cylindrical lens to introduce astigmatism for depth encoding. See Supplementary Sec. A for further details.

Code availability

The code is available online at: https://github.com/ofrigold/pilpel

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References

1. Betzig, E. *et al.* Imaging Intracellular Fluorescent Proteins at Nanometer Resolution. *Science* **313**, 1642–1645 (2006).

2. Rust, M. J., Bates, M. & Zhuang, X. Sub-diffraction-limit imaging by stochastic optical reconstruction microscopy (STORM). *Nature Methods* **3**, 793–796 (2006).

3. Sharonov, A. & Hochstrasser, R. M. Wide-field subdiffraction imaging by accumulated binding of diffusing probes. *Proceedings of the National Academy of Sciences* **103**, 18911–18916 (2006).
4. Sahl, S. J., Hell, S. W. & Jakobs, S. Fluorescence nanoscopy in cell biology. *Nature Reviews Molecular Cell Biology* **18**, 685–701 (2017).
5. Shen, H. *et al.* Single Particle Tracking: From Theory to Biophysical Applications. *Chemical Reviews* **117**, 7331–7376 (2017).
6. Huang, B., Wang, W., Bates, M. & Zhuang, X. Three-Dimensional Super-Resolution Imaging by Stochastic Optical Reconstruction Microscopy. *Science* **319**, 810–813 (2008).
7. Pavani, S. R. P. *et al.* Three-dimensional, single-molecule fluorescence imaging beyond the diffraction limit by using a double-helix point spread function. *Proceedings of the National Academy of Sciences* **106**, 2995–2999 (2009).
8. Shechtman, Y., Sahl, S. J., Backer, A. S. & Moerner, W. E. Optimal Point Spread Function Design for 3D Imaging. *Physical Review Letters* **113**, 133902 (2014).
9. Shechtman, Y., Weiss, L. E., Backer, A. S., Sahl, S. J. & Moerner, W. E. Precise Three-Dimensional Scan-Free Multiple-Particle Tracking over Large Axial Ranges with Tetrapod Point Spread Functions. *Nano Letters* **15**, 4194–4199 (2015).
10. Shechtman, Y., Weiss, L. E., Backer, A. S., Lee, M. Y. & Moerner, W. E. Multicolour localization microscopy by point-spread-function engineering. *Nature Photonics* **10**, 590–594 (2016).
11. Smith, C., Huisman, M., Siemons, M., Grünwald, D. & Stallinga, S. Simultaneous measurement of emission color and 3D position of single molecules. *Optics Express* **24**, 4996 (2016).

12. Backer, A. S. & Moerner, W. E. Single-molecule orientation measurements with a quadrated pupil. *Optics Letters* **38**, 1521–1523 (2013).

13. Hulleman, C. N. *et al.* Simultaneous orientation and 3D localization microscopy with a Vortex point spread function. *Nature Communications* **12**, 5934 (2021).

14. Goldenberg, O., Ferdman, B., Nehme, E., Ezra, Y. S. & Shechtman, Y. Learning Optimal Multicolor PSF Design for 3D Pairwise Distance Estimation. *Intelligent Computing* **2022**, 0004 (2022).

15. Lelek, M. *et al.* Single-molecule localization microscopy. *Nature Reviews Methods Primers* **1**, 39 (2021).

16. Goldenberg, O., Xiao, D. & Shechtman, Y. Deep Learning for Localization Microscopy in 2D and3D. *Accounts of Chemical Research* **59**, 2450–2461 (2026).

17. Nehme, E., Weiss, L. E., Michaeli, T. & Shechtman, Y. Deep-STORM: super-resolution single-molecule microscopy by deep learning. *Optica* **5**, 458 (2018).

18. Nehme, E. *et al.* DeepSTORM3D: dense 3D localization microscopy and PSF design by deep learning. *Nature Methods* **17**, 734–740 (2020).

19. Speiser, A. *et al.* Deep learning enables fast and dense single-molecule localization with high accuracy. *Nature Methods* **18**, 1082–1090 (2021).

20. Hershko, E., Weiss, L. E., Michaeli, T. & Shechtman, Y. Multicolor localization microscopy and point-spread-function engineering by deep learning. *Optics Express* **27**, 6158 (2019).

21. Jouchet, P., Roy, A. R. & Moerner, W. E. Combining deep learning approaches and point spread function engineering for simultaneous 3D position and 3D orientation measurements of fluorescent single molecules. *Opt Commun* **542**, 129589 (2023).

22. Wu, T., Lu, P., Rahman, Md. A., Li, X. & Lew, M. D. Deep-SMOLM: deep learning resolves the 3D orientations and 2D positions of overlapping single molecules with optimal nanoscale resolution. *Opt. Express* **30**, 36761–36780 (2022).

23. Zhang, P. *et al.* Analyzing complex single-molecule emission patterns with deep learning. *Nature Methods* **15**, 913–916 (2018).

24. Nehme, E. *et al.* Learning Optimal Wavefront Shaping for Multi-Channel Imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **43**, 2179–2192 (2021).

25. Sage, D. *et al.* Super-resolution fight club: assessment of 2D and 3D single-molecule localization microscopy software. *Nature Methods* **16**, 387–395 (2019).

26. Archit, A. *et al.* Segment Anything for Microscopy. *Nature Methods* **22**, 579–591 (2025).

27. Gupta, A. *et al.* SubCell: Vision foundation models for microscopy capture single-cell biology. (2024) doi:10.1101/2024.12.06.627299.

28. Ma, C., Tan, W., He, R. & Yan, B. Pretraining a foundation model for generalizable fluorescence microscopy-based image restoration. *Nature Methods* **21**, 1558–1567 (2024).

29. Saguy, A. *et al.* This Microtubule Does Not Exist: Super-Resolution Microscopy Image Generation by a Diffusion Model. *Small Methods* **9**, 2400672 (2025).

30. Baniukiewicz, P., Lutton, E. J., Collier, S. & Bretschneider, T. Generative Adversarial Networks for Augmenting Training Data of Microscopic Cell Images. *Frontiers in Computer Science* **1**, 10 (2019).

31. Goodfellow, I. *et al.* Generative Adversarial Nets. in *Advances in Neural Information Processing Systems* vol. 27 (Curran Associates, Inc., 2014).

32. Isola, P., Zhu, J.-Y., Zhou, T. & Efros, A. A. Image-To-Image Translation With Conditional Adversarial Networks. in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 1125–1134 (2017).

33. Zhu, J.-Y., Park, T., Isola, P. & Efros, A. A. Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. in *Proceedings of the IEEE international conference on computer vision* 2223–2232 (2017).

34. Truedson, A., Gras, K. & Elf, J. The 3D point spread function can be autoencoded in latent space. *bioRxiv* https://doi.org/10.1101/2024.10.14.618231 (2024) doi:10.1101/2024.10.14.618231.

35. Daniel, T. & Tamar, A. Unsupervised Image Representation Learning with Deep Latent Particles. in *International Conference on Machine Learning* 4644–4665 (PMLR, 2022).

36. Daniel, T. & Tamar, A. DDLP: Unsupervised Object-centric Video Prediction with Deep Dynamic Latent Particles. *Transactions on Machine Learning Research* (2024).

37. Opatovski, N. *et al.* Multiplexed PSF engineering for three-dimensional multicolor particle tracking. *Nano Letters* **21**, 5888–5895 (2021).

38. Shalev Ezra, Y. *et al.* High-throughput DNA repair monitoring in Saccharomyces cerevisiae suggests SSB-and DSB-induced chromatin reconfiguration. *Scientific Reports* **15**, 32302 (2025).

39. Saguy, A. *et al.* DBlink: dynamic localization microscopy in super spatiotemporal resolution via deep learning. *Nat Methods* **20**, 1939–1948 (2023).

40. Ovesný, M., Křížek, P., Borkovec, J., Švindrych, Z. & Hagen, G. M. ThunderSTORM: a comprehensive ImageJ plug-in for PALM and STORM data analysis and super-resolution imaging. *Bioinformatics* **30**, 2389–2390 (2014).

41. Orange kedem, R. *et al.* Near index matching enables solid diffractive optical element fabrication via additive manufacturing. *Light: Science & Applications* **12**, 222 (2023).

42. Saguy, A. *et al.* One-click image reconstruction in single-molecule localization microscopy via deep learning. *bioRxiv* https://doi.org/10.1101/2025.04.13.648574 (2025) doi:10.1101/2025.04.13.648574.

43. Backlund, M. P., Joyner, R., Weis, K. & Moerner, W. E. Correlations of three-dimensional motion of chromosomal loci in yeast revealed by the double-helix point spread function microscope. *Molecular Biology of the Cell* **25**, 3619–3629 (2014).

44. Liu, S. *et al.* Universal inverse modeling of point spread functions for SMLM localization and microscope characterization. *Nature Methods* **21**, 1082–1093 (2024).

45. Kingma, D. P. & Welling, M. Auto-Encoding Variational Bayes. (2022) doi:10.48550/arXiv.1312.6114.

46. Haramati, D., Daniel, T. & Tamar, A. Entity-Centric Reinforcement Learning for Object Manipulation from Pixels. in *The Twelfth International Conference on Learning Representations* (2024).

47. Qi, C., Haramati, D., Daniel, T., Tamar, A. & Zhang, A. EC-Diffuser: Multi-Object Manipulation via Entity-Centric Behavior Generation. *arXiv preprint arXiv:2412.18907* (2024).

48. Ferdman, B. *et al.* VIPR: vectorial implementation of phase retrieval for fast and accurate microscopic pixel-wise pupil estimation. *Optics Express* **28**, 10179 (2020).

49. Karempudi, P. *et al.* Three-dimensional localization and tracking of chromosomal loci throughout the Escherichia coli cell cycle. *Communications Biology* **7**, 1443 (2024).

Self-supervised generation of realistic training data enables nanoscale localization in challenging conditions

Supplementary Material

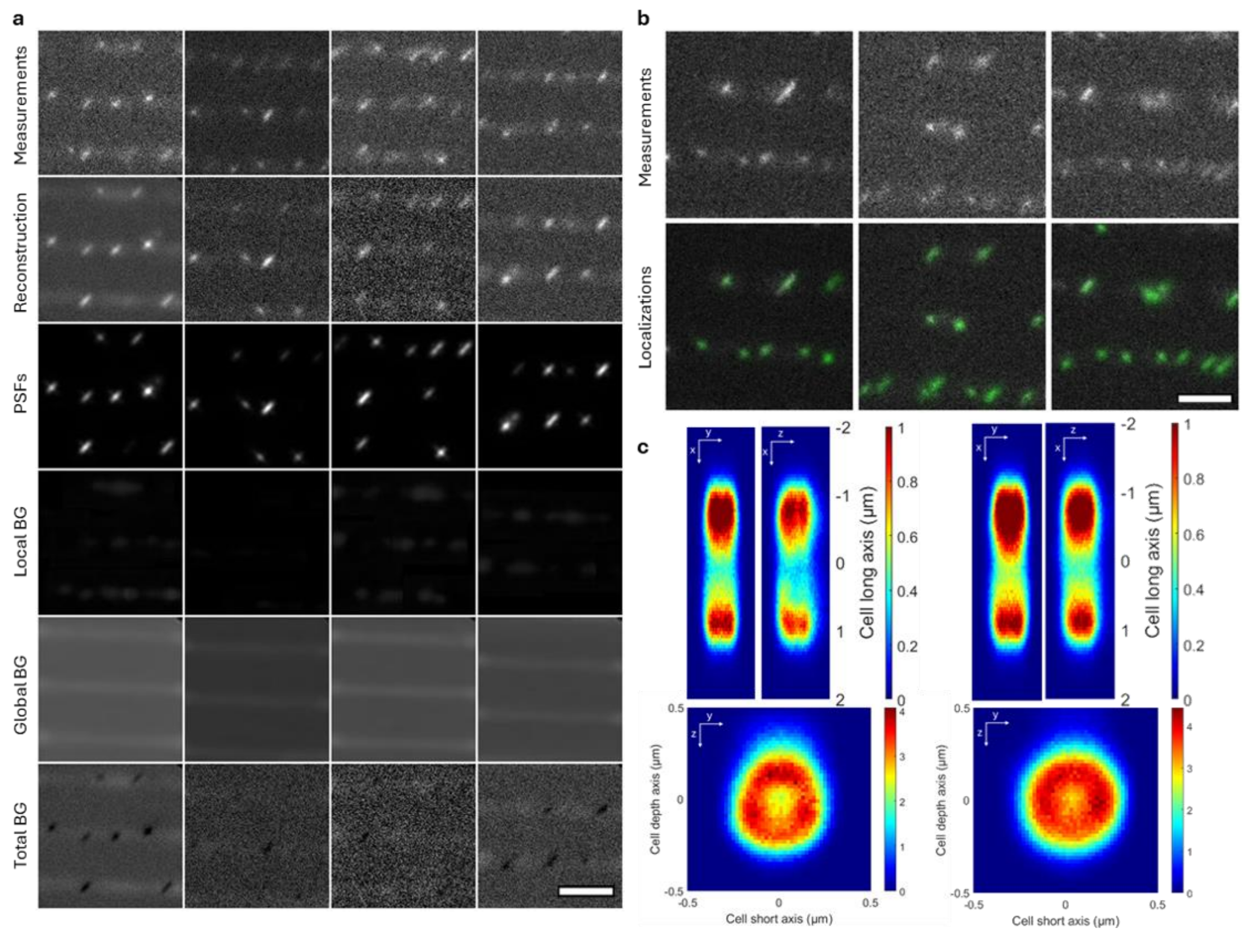
A. 3D localization in bacteria with astigmatism PSF

To further validate the versatility of our method across different biological systems and PSF-engineering approaches, we applied PILPEL to 3D localization data from bacteria using astigmatism-based depth encoding. This dataset, published by Karempudi et al.[1], involves imaging of fluorescently labeled chromosomal loci in live *Escherichia coli* cells. In this experiment, three chromosomal loci were labeled with yellow fluorescent protein (YFP) for imaging. The cells were imaged using a 100x/1.45NA oil immersion objective in a widefield microscopy setup equipped with a cylindrical lens to introduce astigmatism for 3D localization (see Ref[1] for details). Similar to the Tetrapod PSF used in previous experiments, astigmatism also encodes axial position, but through the ellipticity and orientation of the PSF, which depend on the distance from the focal plane. The astigmatic PSF was experimentally calibrated using fluorescent beads and phase retrieval via VIPR[2]. Cells were grown and imaged in a microfluidic device, with each field of view containing 20 growth channels.

*Supplementary Figure 1*a illustrates PILPEL's learned decomposition of the bacterial imaging data. The first row shows five representative measured images, each containing multiple bacterial cells with visible fluorescent loci. The second row displays PILPEL's full reconstructions. The remaining rows show the PSFs and the non-PSF components. Notably, the background components reveal a distinct, structured signal corresponding to the autofluorescence background of the cells in the microfluidic channels, learned automatically by the model.

Using the PILPEL-generated labeled dataset, we trained DS3D[3] for 3D localization on the astigmatic data. *Supplementary Figure 1*b shows an example of the 3D localization rendering, displayed as green-on-white overlap. To assess localization accuracy in the absence of absolute ground truth, we compared the detected positions to the 3D localizations produced by the FD-DeepLoc pipeline reported in Ref[1]; observing the same hallmark spatial patterns: our reconstructions reproduced the ring-shaped yz-plane distribution of the *oriC* locus and the expected long and short-axis localizations (*Supplementary Figure 1*c). Quantitatively, the mean absolute error between our localizations and those from the original paper for the *oriC* data is $\sim$ 17 nm in X and Y axes and $\sim$ 58 nm in Z axis, over 128K localizations.

This bacterial localization experiment is significant for several reasons. First, it demonstrates again that our approach readily adapts to different PSF engineering strategies - simply by replacing the physical PSF model in the decoder, without any other architectural changes. Second, this biological system exhibits different background patterns and autofluorescence characteristics compared to the yeast cells. Notably, in order to train a localization network on their data, the original work[1] required a substantial simulation pipeline, involving cell segmentation masks of phase-contrast images, extraction of intensity profiles, and careful modeling of cell-specific background fluorescence patterns to create training data resembling the real measurements. In contrast, our PILPEL model learns these complex background characteristics and noise patterns directly from the experimental data without any manual intervention or explicit modeling of the bacterial cell morphology.



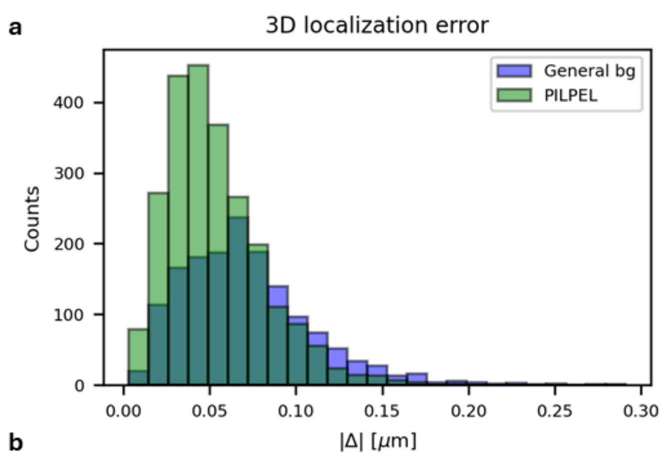
Supplementary Figure 1: 3D localization of loci in E. coli. (a) PILPEL output components. The first row shows five measured images; the second row shows PILPEL's reconstruction; the remaining rows show the reconstruction components. The Local BG and Global BG rows display the distinct background corresponding to the cells in the microfluidic channels, learned by the model. (b) 3D localization rendering by DS3D (DeepSTORM3D) trained on PILPEL-generated data. (c) 3D spatial distribution of the oriC locus in cell-internal coordinates. The data are two-dimensional histograms of localized emitter positions, pooled from time-lapse imaging across all detected cell areas. Results on the left are from the original paper, and results on the right are ours, demonstrating a highly similar localization pattern. The top panel shows the distribution along the cell's long (x) and short (y) axes, and the bottom panel shows the radial distribution in the short (y) and depth (z) axes (yz-plane). Scale bar, 2 μm.

B. Controlled simulation for performance assessment

To generate diverse and challenging noise patterns for the `PerlinData` test set, our implementation uses Perlin noise scale parameters ranging from 0.5 to 1.7 and intensity values from 5 to 14, which control the spatial frequency and amplitude of the noise, respectively. These parameters were chosen to produce diverse patterns—noisy enough to challenge localization accuracy, but not so severe as to obscure the PSF signal. The results presented here provide a detailed analysis of the performance gains referenced in the main text.

The histogram in *Supplementary Figure 2*a compares the 3D localization error distributions for DS3D trained on the "General bg" versus the "PILPEL" dataset. Visually, the PILPEL-trained model's distribution (green) is clearly shifted toward lower errors, with a higher peak and a smaller high-error tail than the original model (blue). This improvement is quantified by the mean absolute error, which dropped from 0.071 µm for the original model to 0.053 µm for the PILPEL-trained model.

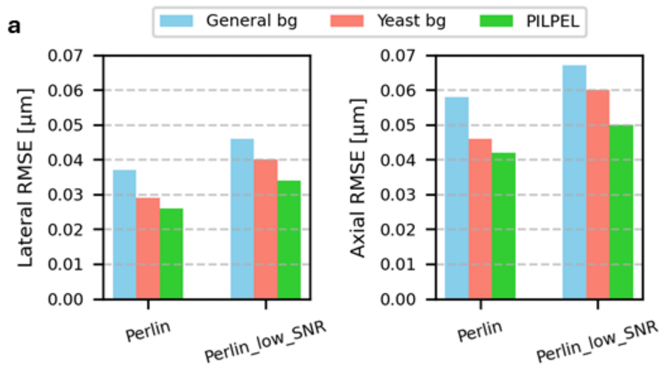
The table in *Supplementary Figure 2*b further details these robustness gains, breaking down the performance (ΔRMSE and ΔJacc) when comparing the PILPEL-trained model against two different baselines: the standard "General bg" simulator and the "Yeast bg" simulator (which included manual prior knowledge). When compared to the "General bg" baseline, the PILPEL model improved ΔRMSE by -19 nm (-28%) and ΔJacc by $+0.47$ on the `Perlin` test set. On the more difficult `Perlin Low SNR` set, the error reduction was -21 nm (-26%) with a ΔJacc gain of $+0.21$. Critically, PILPEL also outperformed the model trained with manually added prior knowledge ("Yeast bg"). Against this stronger baseline, PILPEL was still better by -12 nm (-16%) ΔRMSE and $+0.22$ ΔJacc on the challenging `Perlin Low SNR` set. Lateral and axial RMSE results are presented in *Supplementary Figure 3*. These results confirm that PILPEL's learned generative approach is more effective than training on manually-designed synthetic datasets, with the large Jaccard index gains demonstrating a higher rate of correct emitter detection.



b

	PILPEL vs. General bg			PILPEL vs. Yeast bg		
Test set	ΔRMSE		Δ Jacc	ΔRMSE		Δ Jacc
Perlin	-19 [nm]	**-28 [%]**	**+0.47**	-5 [nm]	-9 [%]	+0.40
Perlin Low SNR	-21 [nm]	**-26 [%]**	**+0.21**	-12 [nm]	-16 [%]	+0.22

Supplementary Figure 2: Robustness experiment on simulations– DS3D localization performance. (a) 3D localization error comparison: DS3D trained on the generated dataset vs. DS3D trained on the standard simulated dataset. Mean absolute error for the generated model is 0.053 µm, *compared to 0.071 µm for the original. (b) Improvement of training on generated data. Localization RMSE and Jaccard index for DS3D trained on the generated dataset vs. training on yeast-BG and original-DS3D datasets.*



b

	PILPEL vs. General bg				PILPEL vs. Yeast bg			
Test set	ΔLateralRMSE		ΔAxialRMSE		ΔLateralRMSE		ΔAxialRMSE	
Perlin	-11 [nm]	-29 [%]	-16 [nm]	-27 [%]	-3 [nm]	-10 [%]	-4 [nm]	-8 [%]
Perlin Low SNR	-12 [nm]	-26 [%]	-17 [nm]	-25 [%]	-6 [nm]	-15 [%]	-10 [nm]	-17 [%]

Supplementary Figure 3: Lateral and axial localization accuracy comparison. Quantitative evaluation of the same experiment shown in Supplementary Figure 2, with the RMSE separated into lateral (x, y) and axial (z) components. Results are reported for all evaluated methods and datasets, demonstrating consistent improvement of our approach in both dimensions.

C. Experimental evaluation of super-resolution reconstructions

To evaluate PILPEL's improvement on the experimental data lacking an absolute ground truth, we analyzed the localization results and super-resolution reconstructions of the 2D and 3D STORM experiments, applying complementary analyses to the two resulting datasets: standard DS/DS3D localizations and PILPEL-trained localizations. PILPEL detects more localizations than the original network on the same frames, and qualitative comparisons demonstrate that PILPEL produces a denser, fuller, and more continuous reconstruction; the analyses below verify that this increase is mostly enhancing true biological structures.

First, to validate that the increase in localizations yields a more complete structural representation without introducing artifacts, we performed a line-profile analysis on the 2D STORM data of microtubules, showing that the signal increase produced by PILPEL is spatially concentrated at the microtubule filaments. Across three microtubule ROIs, we compared width-averaged intensity profiles on the filament ("center") and in two adjacent structure-free regions ("left"/"right") for DS and PILPEL (*Supplementary Figure 4*a). On the microtubule, PILPEL raised the mean profile intensity by 0.039, 0.087, and 0.117 a.u. over DS across the three ROIs, whereas in the empty regions the increase stayed at the background floor (≤0.006 a.u.). Thus, while false positives can occur, the extra localizations PILPEL recovers are predominantly associated with genuine microtubule signal.

Similarly on the 3D STORM of mitochondria, we performed density analysis on 20 regions of interest (ROIs) containing distinct mitochondrial structures and 20 control ROIs situated in empty background regions (*Supplementary Figure 5*a). We defined 20 ROIs on mitochondrial tubules, each a rectangular box in *x–y* restricted to the axial slab containing that tubule. Each ROI was translated laterally onto an adjacent structure-free region, preserving box dimensions and *z*-slab. This yields 20 matched pairs differing only in the presence of a mitochondrion— identical volumes, axial position, and confidence threshold (40) for both methods (*Supplementary Figure 5*b). Densities are per unit volume (loc/μm^3) with total volume of 11.18 μm^3 per group. The added density is 635.5 loc/μm^3 on structure and 4.11 loc/μm^3 off structure (see *Table 1*), and the median nearest-neighbor distance on the structure-areas was finer in all 20 ROIs (*Supplementary Figure 5*e).

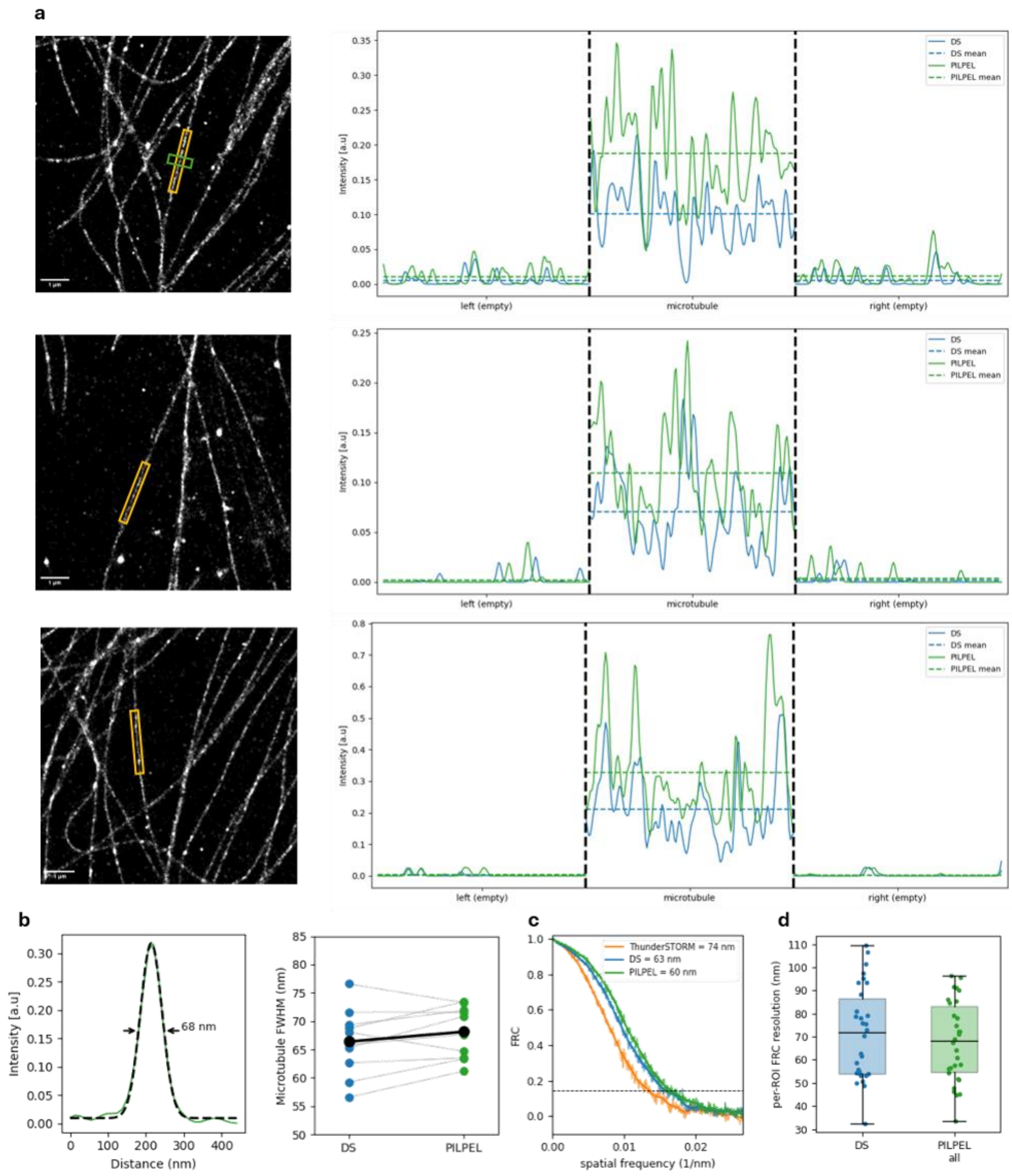
Table 1: Localization counts and densities in structure and background ROIs for DS3D and PILPEL-trained networks. Pooled over all ROIs, n = 20 matched ROI pairs, total volume 11.18 μm^3 per group.

	DS3D	**PILPEL**
On-structure localizations	3,329 (297.9 loc/μm^3)	10,432 (933.4 loc/μm^3)
Off-structure localizations	5 (0.45 loc/μm^3)	51 (4.56 loc/μm^3)

In a second analysis for biological structure validation, we measured the apparent filament width from cross-sectional profiles of isolated microtubules (*Supplementary Figure 4*b). Gaussian fits yielded a mean FWHM of 66.4 ± 5.9 nm for DS and 68.1 ± 4.6 nm for PILPEL (consistent with the apparent width expected for antibody-labeled microtubules[4]), with no significant difference between the methods. Critically, PILPEL's higher detection rate adds signal along the microtubules without artificially broadening them. In the 3D data, an axial cross-section through a mitochondrial tubule resolves the hollow cross-section, matching the expected mitochondrial geometry (*Supplementary Figure 5*c and d).

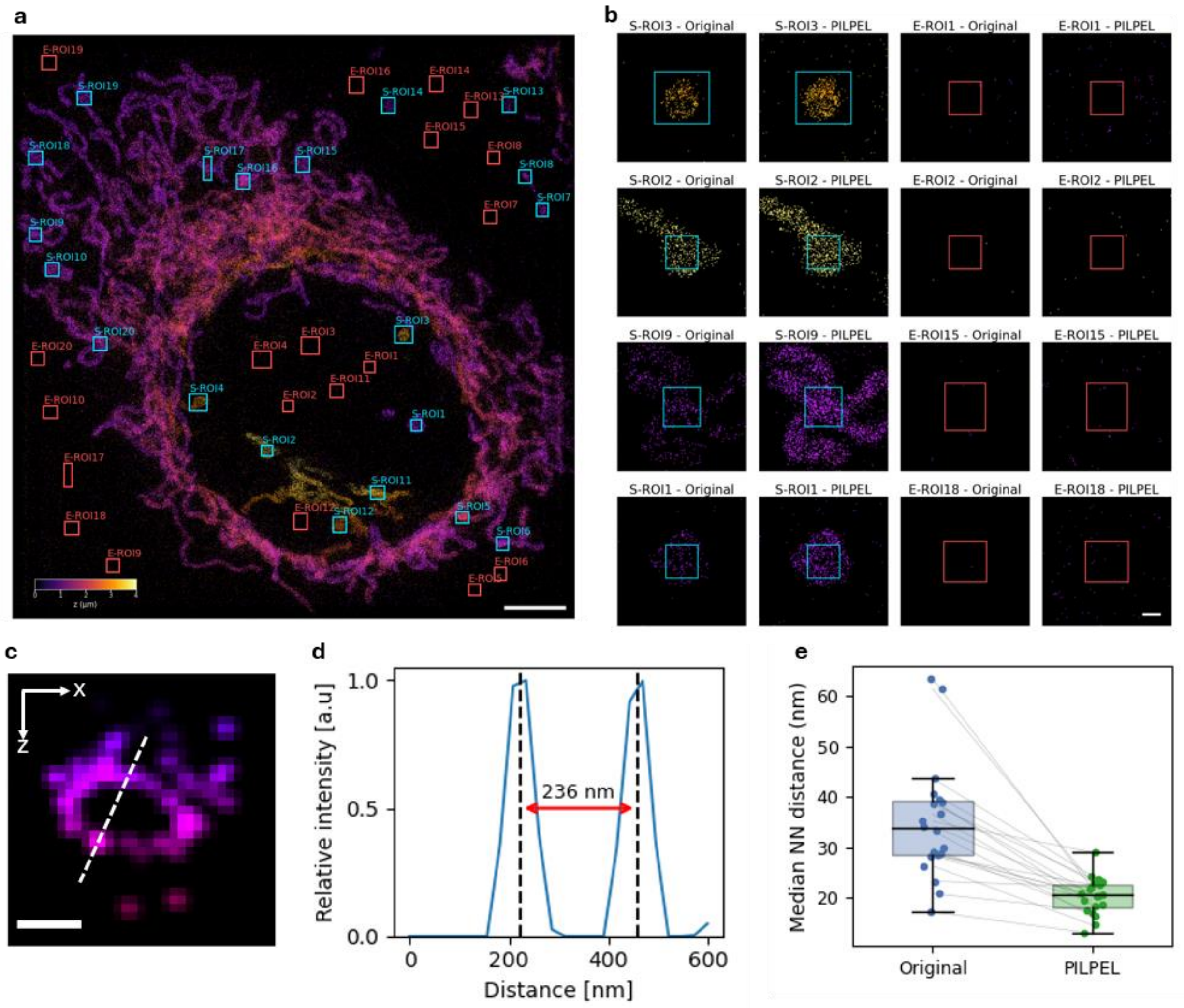
Third, we evaluated the reconstructions empirically using Fourier Ring Correlation (FRC)[5] on independent even/odd frame splits of the microtubule dataset (*Supplementary Figure 4*c). Stochastic hallucinated detections would not correlate between the two halves and would lower the measured resolution. At the full field-of-view (all counts), the methods yielded comparable global resolutions (DS 63 nm, PILPEL 60 nm), with PILPEL carrying approximately twice the localizations. In addition to the full field, we repeated the FRC analysis in fixed 5 µm tiles across the field of view (n = 32; *Supplementary Figure 4*d). Consistent with the global result, PILPEL was comparable to DS per tile, with a modest median improvement of ~4 nm.

Together, these results indicate that PILPEL's additional localizations are consistent with genuine biological structure and mostly confined to regions occupied by structure.



Supplementary Figure 4: Validation of 2D STORM reconstructions. (a) Super-resolution reconstructions of microtubule filaments and corresponding intensity line profiles across selected regions of interest (yellow ROIs). The profiles compare the original DeepSTORM (DS) localizations with PILPEL-trained localizations, showing that the signal increase is concentrated at the genuine microtubule structure (center zone) and remains at the near-zero background floor in adjacent structure-free regions (left/right zones).(b) Validation of apparent filament width. Left: A representative Gaussian cross-section fit of an isolated microtubule (green box, top row of (a)).

Right: Paired comparison of FWHM measurements across n=10 ROIs. PILPEL's higher detection rate preserves the apparent filament width without artificial broadening (mean FWHM: DS 66.4 ± 5.9 nm, PILPEL 68.1 ± 4.6 nm). (c) Global Fourier Ring Correlation (FRC) evaluating resolution on independent even/odd frame splits, where resolution is the inverse of the spatial frequency at which the FRC curve crosses the 1/7 threshold (dashed line). ThunderSTORM: 74 nm, DS: 63 nm, PILPEL: 60 nm. (d) Per-ROI FRC resolution across fixed 5 µm tiled regions (n = 32). PILPEL is comparable to DS with slight improvement while detecting 1.96× more localizations.

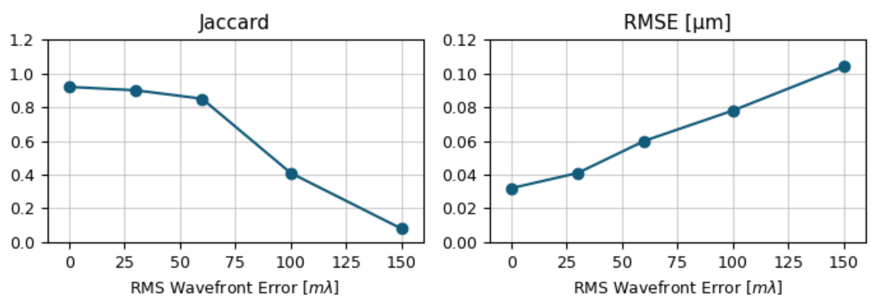


Supplementary Figure 5: Volumetric density and structural validation of 3D STORM mitochondria reconstructions. (a) Full field-of-view 3D STORM reconstruction color-coded by axial depth (z). For density evaluation, 20 on-structure regions of interest (S-ROIs, cyan boxes) were defined on mitochondrial tubules, and 20 volume-matched off-structure background regions (E-ROIs, red boxes) were selected in close empty areas. Scale bar: 5 µm. (b) Representative x-y projections of on-structure and off-structure ROIs from (a), comparing the Original (DS3D) and PILPEL-trained reconstructions. Scale bar: 0.5 µm (c) Axial cross-section (x-z plane) of a PILPEL-reconstructed mitochondrial tubule, successfully resolving the expected hollow biological structure. Scale bar: 0.2 µm. (d) Intensity profile measured along the dashed line in (c).

The distance between the two peaks yields an apparent tubule diameter of 236 nm. (e) Paired comparison of the median nearest-neighbor (NN) distance across the 20 on-structure ROIs. PILPEL consistently yields a finer median NN distance than the baseline across all measured regions.

D. Robustness to PSF aberrations

We tested robustness to PSF mismatch by perturbing the VIPR phase mask with a fixed random Zernike combination (astigmatism, coma, spherical) at RMS wavefront error amplitudes ranging from a perfect match (0 mλ) to severe mismatch (150 mλ). PILPEL was retrained from scratch on PerlinData with each miscalibrated prior (used in both the initial proposal module and the physical decoder, as in the usual pipeline), then a fresh DS3D was trained on its generated images. On the test set Jaccard remains high within the diffraction-limited regime (J = 0.9, RMSE = 41 nm at 30 mλ; J = 0.85, RMSE = 60 nm at 60 mλ), degrades sharply once mismatch exceeds the Maréchal threshold (≈71 mλ), reaching J = 0.41, RMSE = 80 nm at 100 mλ, and collapses at 150 mλ (J = 0.08) under extreme miscalibration (see *Supplementary Figure 6*). These results quantify the prior-PSF dependence noted in the limitations section: PILPEL tolerates miscalibration within the diffraction-limited regime, while larger mismatch beyond the Maréchal threshold degrades performance. Future work could remedy this limitation by introducing an additional latent variable to learn and model pupil function aberrations.



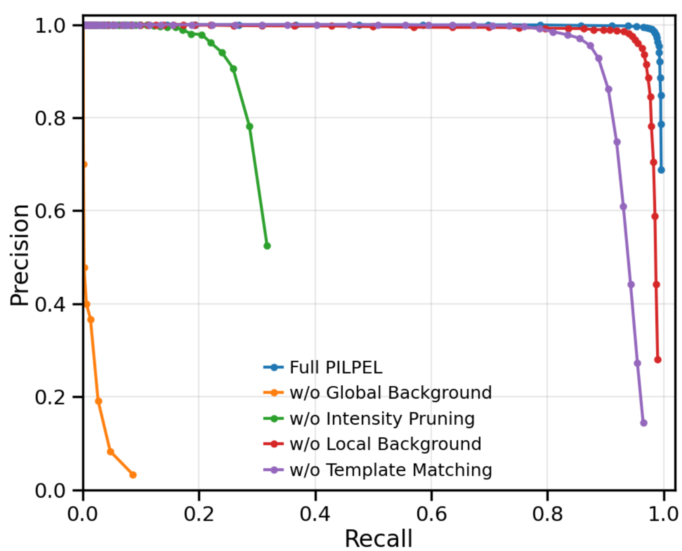
Supplementary Figure 6: Aberration robustness. Downstream localization performance (Jaccard and RMSE) of DS3D trained on PILPEL-generated datasets, as the PSF model is increasingly miscalibrated.

E. Ablation Study

To quantify the contribution of each component in the PILPEL framework, we evaluated the localization performance on the `PerlinData` test set under different configurations. *Supplementary Figure 7* presents the Precision-Recall curves for all variants, and Table 2 summarizes the key metrics, calculated at a working confidence threshold of 100. The results demonstrate that explicitly modeling the background is critical. Removing the **Global Background** module leads to a collapse in performance (recall < 1%). This occurs primarily because artifacts in the generated images cause a large deviation from the test set; consequently, most PSFs are drowned to zero as the local object components attempt to compensate and learn the general image structure. The **Local Background** is also important for reconstruction, and its removal degrades performance, though the degradation is less severe, likely due to the presence of the global background. Removing the **intensity-pruning** mechanism prevents the model from zeroing out spurious proposals. Since all initialized particles remain active at fixed high intensity, the generated training images are flooded with emitters at random locations, producing unrealistically dense and cluttered training data, causing recall to be capped at ~0.3 regardless of detection threshold, as seen in *Supplementary Figure 7*. While this configuration yields slightly higher precision (0.97) among all variants, it comes with a drastic drop in sensitivity, and the Jaccard index (0.21) correctly captures the severity of this failure. Finally, omitting the **template-matching** initialization results in unreliable particle location proposals, leading to an excess of false detections and degraded precision throughout, which reduces the Jaccard index to 0.75. The full PILPEL configuration yields the highest overall performance as measured by the Jaccard index (0.92) and the lowest localization error (RMSE 0.032 µm), confirming the necessity of all proposed modules.

Table 2: Ablation Study Results

Configuration	Precision (↑)	Recall (↑)	Jaccard (↑)	RMSE (µm) (↓)
Full PILPEL	0.93	**0.99**	**0.92**	**0.032**
w/o Local Background	0.81	0.98	0.80	0.035
w/o Global Background	0.38	0.005	0.005	0.054
w/o intensity-pruning	**0.97**	0.21	0.21	0.051
w/o template-matching	0.8	0.91	0.75	0.034



Supplementary Figure 7: Ablation Study: Precision-Recall Curves. Performance comparison of the full PILPEL model (blue) against configurations with individual modules removed. The full model yields the most robust detection envelope.

F. Computational details

All PILPEL training runs were performed on a single NVIDIA Titan RTX GPU (32 GB) using PyTorch. We trained the model for 50 epochs with a batch size of 4 using Adam optimizer, and a learning rate of 1e-5. We held out 10% of each dataset for validation. The reconstruction loss was either pixel-wise MSE or a VGG-based perceptual loss[6]. Based on our empirical observations across datasets, MSE appears preferable for noisy backgrounds with minimal structure (blinking emitters), while perceptual loss better suits textured backgrounds with defined patterns (yeast, Perlin noise); however, a systematic evaluation across more datasets is required to draw definitive conclusions.

For each dataset, we performed VIPR[2] phase retrieval on calibration-bead measurements to estimate a phase mask with experiment-specific optical parameters and the expected axial range (see *Supplementary Figure 10*). The resulting PSF model served both as the differentiable PSF generator in the decoder and as the basis for a PSF dictionary used by the correlation-based prior for initial coordinate proposals. We discretized the PSF into 21 z-planes across the modeled axial range. To generate the proposals, we partitioned the image into a uniform grid of patches, computed the 3D cross-correlation of each patch with the dictionary, and re-centered the patches around the coordinates of maximum correlation. The patch size is determined by the PSF model dimensions.

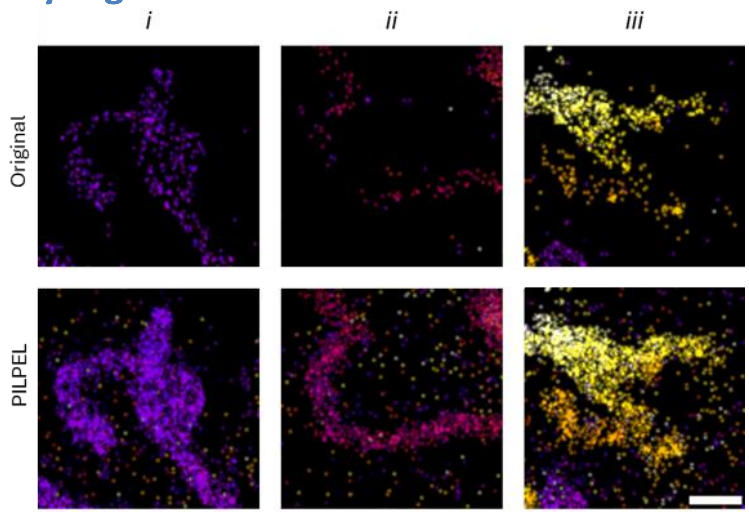
Number of particles to encode was chosen per dataset (detailed in Table 3), while high initialization used for typically dense emitters, and reduced if the data was substantially sparse, trading efficiency against potential missed localizations. The latent dimension was 10 for the local-background features and 20 for the global-background encoding. Architecture, latent dimensions and all components not modified in our method followed DLP[7], using the default recommended hyper-parameters from the open-source code. Training typically converged within ~8 hours on average, depending on the dataset and image size. After training, the decoder served as a generator to synthesize 10,000 labeled images for DS3D training. PILPEL inference (i.e. data generation) was performed on the same hardware at approximately 5 frames/s, depending on image size.

Dataset-specific PILPEL configurations and training details appear in Table 3.

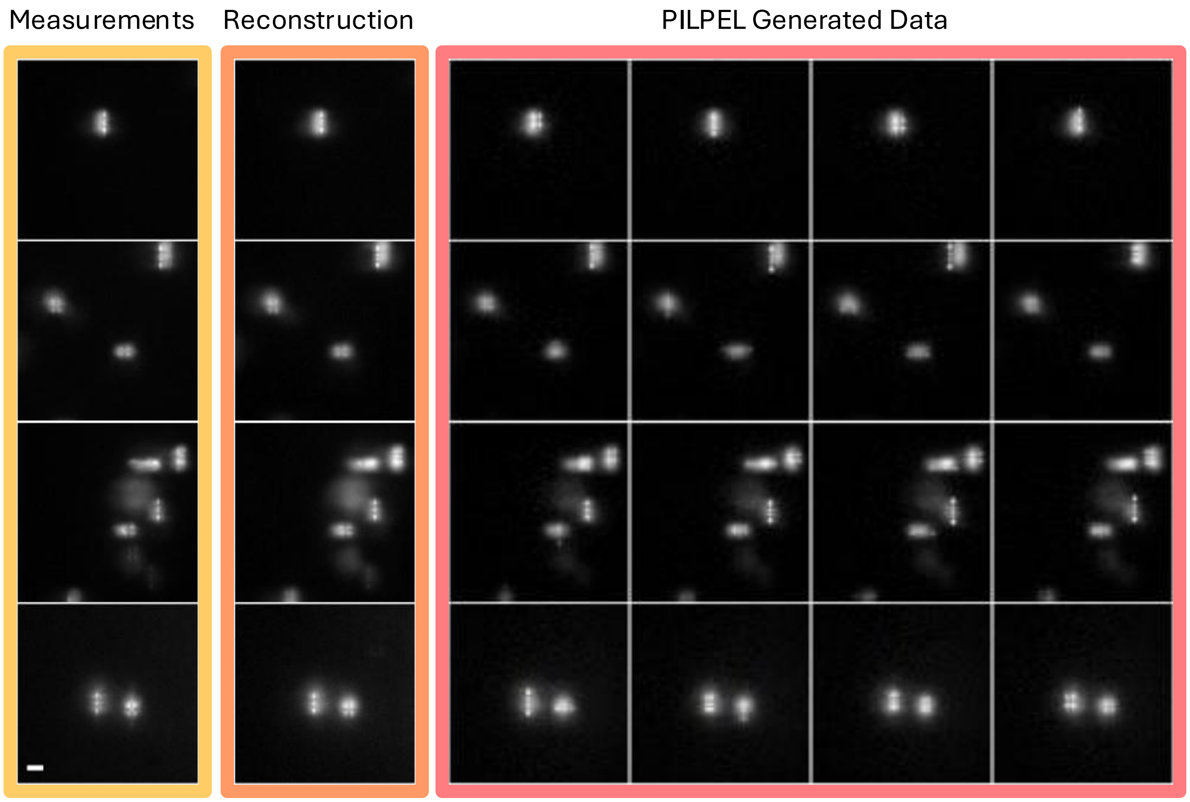
Table 3: PILPEL hyperparameters per dataset

Parameter	Perlin	3D STORM	Yeast	2D STORM	Bacteria
Learning rate	1e-4	1e-5	1e-4	1e-5	1e-5
Reconstruction loss type	VGG	MSE	VGG	MSE	MSE
Image size (px)	193	129	193	65	129
Patch size (px)	32	32	32	16	32
Initial latent particles #	60	16	60	36	16
Training set size (images)	1000	3000	1200	4200	4000
Training time (min/epoch)	7.70	6.57	9.72	11.76	14.46
Total runtime (h)	6.4	5.5	8.1	9.8	12
Inference time (images/min)	155	571	180	409	380

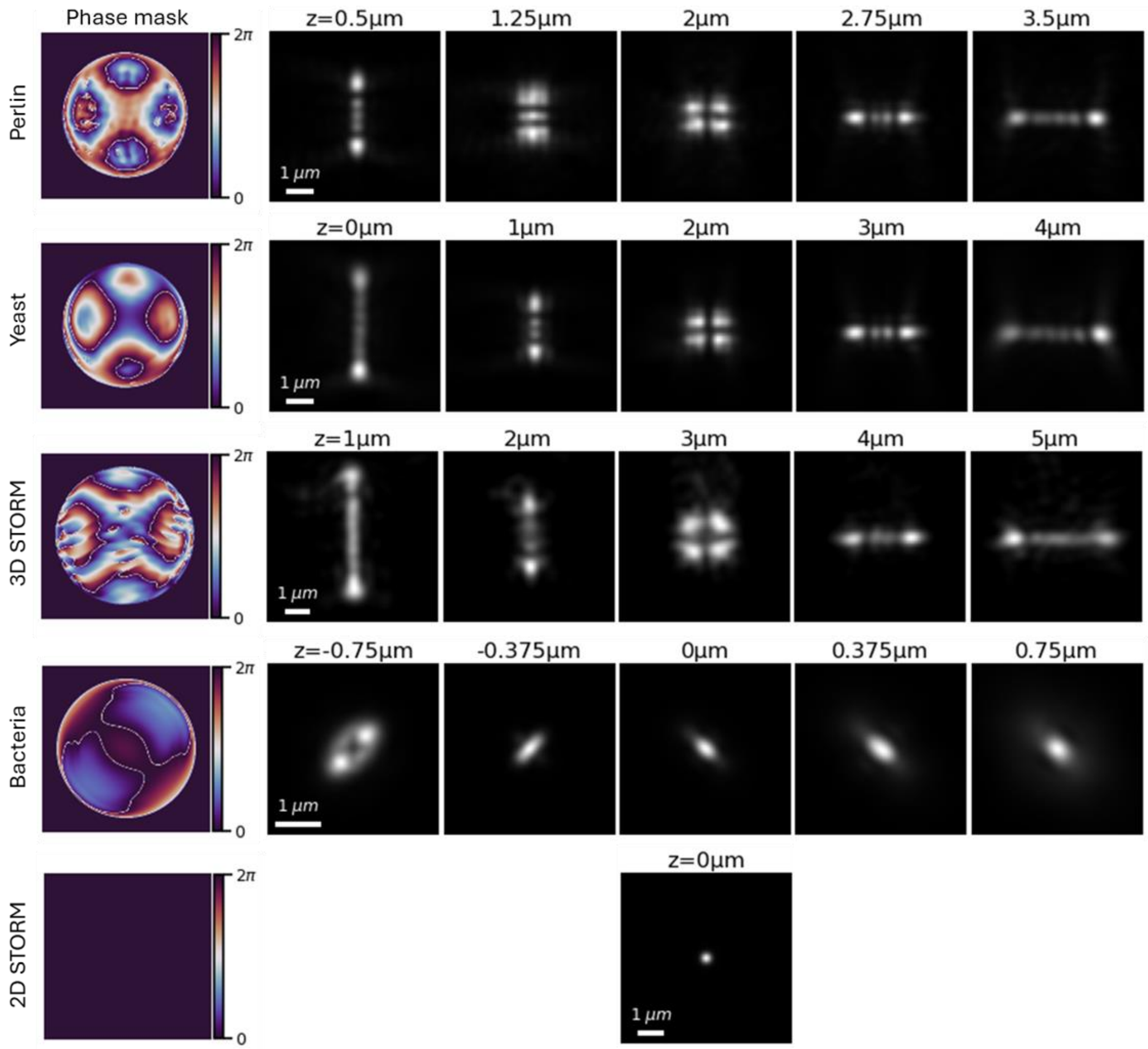
G. Supplementary Figures



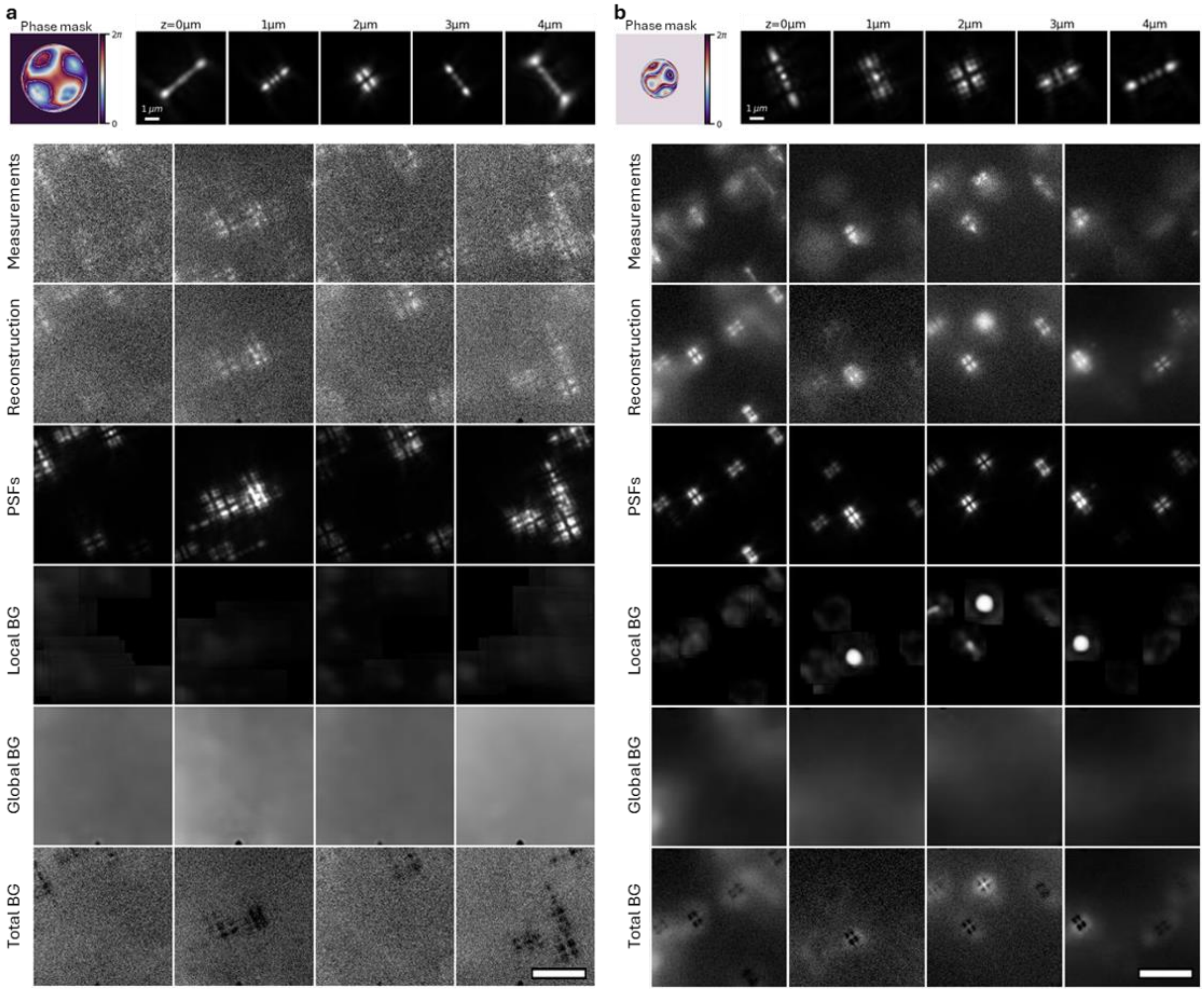
Supplementary Figure 8: Unfiltered 3D super-resolution reconstruction of mitochondria. Magnified views of regions i, ii, and iii from Figure 5 in the main text. Top row: reconstruction using DS3D trained on standard simulated data. Bottom row: reconstruction using DS3D trained on PILPEL-generated data. As discussed in the main text, the PILPEL-trained reconstruction shown in Figure 5 was post-processed with a density filter to facilitate clearer visual comparison. Here, we show the raw, unfiltered PILPEL-trained reconstruction. As can be seen, the higher number of detections enables more complete structural recovery, which in turn allows such filtering to enhance visualization. In contrast, the standard-trained reconstruction is more sparse and applying the same filter would erase the remaining structural information. Scale bar, 1 μm.



Supplementary Figure 9: Realistic labeled training data generation. Measured images are passed through the encoder-decoder model to generate the reconstructions. To create a labeled dataset, the encoded PSF attributes are randomly perturbed, and the stochastic latent variables are resampled. This process produces realistic image variants that resemble the original measurements. Since the PSF coordinates fed into the generator are known, the resulting dataset is fully labeled and can be used to train a 3D localization network. Scale bar, 2 µm



Supplementary Figure 10: PSF models used across experiments. Phase masks retrieved with VIPR from calibration-bead measurements and their corresponding PSF models, illustrating the depth-dependent PSF shapes used for reconstruction and data generation. Shown are Tetrapod PSFs (used in the synthetic Perlin dataset, 3D STORM of mitochondria and yeast DNA loci), an astigmatic PSF for E. coli experiment, and the standard diffraction-limited PSF used in the 2D STORM experiment



Supplementary Figure 11: Additional PILPEL reconstructions and component decomposition of experimental data. Left: 3D STORM of mitochondria labeled with Alexa Fluor 647, and imaged with a rotated Tetrapod designed for $\lambda = 670\ nm$ *and an axial range of* $4\ \mu m$*. Right: fluorescent DNA loci in yeast cells, labeled with eGFP and imaged with a rotated Tetrapod designed for* $\lambda = 515\ nm$ *and an axial range of* $4\ \mu m$*. Top insets show the phase masks and the calibrated PSF model retrieved with VIPR. Scale bars, 1 µm.*

References

1. Karempudi, P. *et al.* Three-dimensional localization and tracking of chromosomal loci throughout the Escherichia coli cell cycle. *Communications Biology* **7**, 1443 (2024).
2. Ferdman, B. *et al.* VIPR: vectorial implementation of phase retrieval for fast and accurate microscopic pixel-wise pupil estimation. *Optics Express* **28**, 10179 (2020).
3. Nehme, E. *et al.* DeepSTORM3D: dense 3D localization microscopy and PSF design by deep learning. *Nature Methods* **17**, 734–740 (2020).
4. Mikhaylova, M. *et al.* Resolving bundled microtubules using anti-tubulin nanobodies. *Nat Commun* **6**, 7933 (2015).
5. Nieuwenhuizen, R. P. J. *et al.* Measuring image resolution in optical nanoscopy. *Nat Methods* **10**, 557–562 (2013).
6. Hoshen, Y., Li, K. & Malik, J. Non-adversarial image synthesis with generative latent nearest neighbors. in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* 5811–5819 (2019).
7. Daniel, T. & Tamar, A. DDLP: Unsupervised Object-centric Video Prediction with Deep Dynamic Latent Particles. *Transactions on Machine Learning Research* (2024).